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Transformer-Based Neural Surrogates for Microbial Metabolic Modeling

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Tiivistelmä — Referat — Abstract

Flux Balance Analysis (FBA) predicts how metabolic fluxes in microorganisms change with nutrient conditions, but each prediction requires solving a dedicated optimization problem and the nutrient uptake bounds needed for FBA are often not available in experimental data. These limitations make FBA difficult to use directly as a component in larger systems, such as machine learning pipelines. In this thesis, we develop FluxTransformer, a neural surrogate that maps nutrient uptake bounds to complete signed flux distributions. The model was trained using data generated by FBA or pFBA from the *Escherichia coli* core model, iML1515, and the *Saccharomyces cerevisiae* model Yeast9. The evaluation included simulated flux prediction, learned reaction representations, training-efficiency experiments and transfer to experimental tasks.

On simulated data, FluxTransformer predicts biomass and major fluxes accurately for all three metabolic models, with pooled R^2 above 0.9 in each case. The learned representations retain reaction identity and show partial alignment with pathway membership and flux behavior. In the experimental growth task, we outperform the previously proposed Artificial Metabolic Network (AMN) variants and the TabPFN baseline. In the missing-flux task, we achieve performance similar to the previously proposed Metabolic-Informed Neural Network (MINN).

FluxTransformer demonstrates that Transformer-based surrogates can approximate FBA predictions in increasingly large metabolic models. Its application to experimental tasks using frozen pretrained weights also indicates that simulated pretraining can generate reusable metabolic priors. These results support further development toward more general metabolic surrogates.

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The source code is available at: https://github.com/ruokokoski/metabolic-NN

Essentially, all models are wrong, but some are useful.
— George E. P. Box

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1. Introduction

Microbial cells are small biochemical factories. They take in nutrients from their surroundings and process them through thousands of biochemical reactions to produce biomass, energy, and valuable products [52]. Their ability to use diverse nutrients makes microorganisms promising for circular economy applications, where industrial and agricultural side streams could serve as growth substrates. Many of these streams contain carbon, nitrogen, and other nutrients that microbes need to grow. If these resources can be efficiently converted into biomass or proteins, they offer an opportunity to create value from materials that would otherwise go to waste [32]. Figure 1.1 shows this process, from nutrient uptake to the production of microbial biomass and secreted metabolites.

How efficiently microorganisms convert these nutrients depends strongly on their environment. Changes in nutrient availability can alter growth, pathway activity, and product yield [69]. Understanding and predicting these responses is important for choosing the right organisms, designing growth media, and reducing the number of costly experiments needed to optimize production [21].

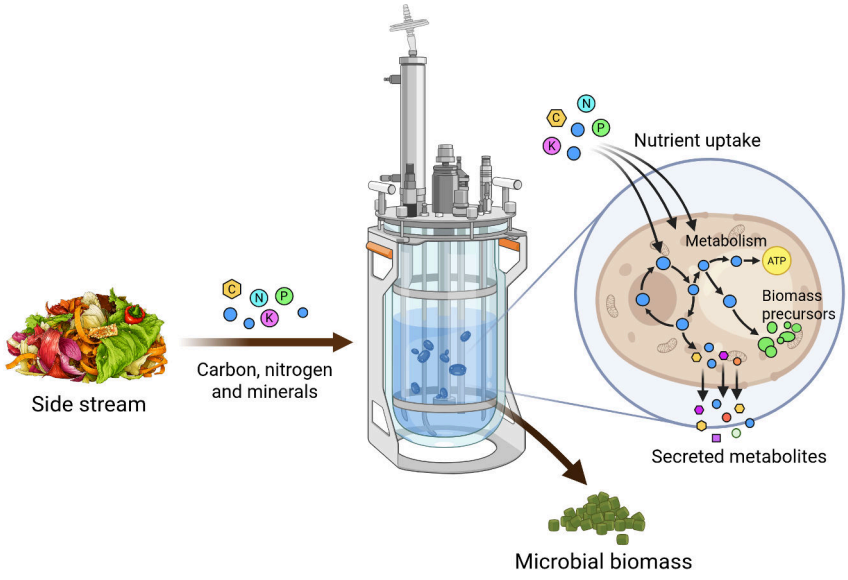


Figure 1.1: Microorganisms take up nutrients from a side stream and convert them into biomass and secreted metabolites.

Metabolic fluxes are one way to describe this behavior. They measure how fast reactions occur in a metabolic network [5]. Fluxes reflect how the cell functions: which pathways are active, how nutrients are allocated throughout metabolism, and how cellular resources are directed toward growth or product synthesis [52]. Growth is a measurable outcome of this metabolic activity and can be compared directly with experimental data. Examining growth alongside the other predicted fluxes helps to evaluate whether a metabolic model responds to different nutritional environments in a biologically meaningful way.

Flux Balance Analysis (FBA) is a common method for predicting metabolic fluxes using metabolic models. It represents metabolism through stoichiometric constraints and predicts a flux distribution by solving an optimization problem, typically with biomass production as the objective [54]. FBA simplifies metabolism into a steady flow problem, where nutrients enter the network, reactions convert and redistribute them, and the model requires that metabolites are produced and consumed at equal rates. This compact formulation leaves out several characteristics of real cells, including metabolite concentrations, regulatory effects, and changes over time [52].

One specific challenge of FBA is that it relies on knowledge of uptake constraints, which specify limits on how rapidly nutrients can be taken up from the environment. The medium composition defines which nutrients are available, but does not directly determine the uptake rates required by the model. These constraints are therefore a source of uncertainty, and their specification can affect the resulting phenotype predictions [48].

Another challenge in using FBA is that each nutrient condition requires solving a separate optimization problem. It would be useful to adjust nutrient conditions based on how these changes affect predicted growth or product formation, for example when designing growth media [34, 50]. Gradient-based methods can use information on how a small change in each nutrient affects the predicted outcome. However, conventional FBA solvers do not support backpropagation through the optimization step, which makes FBA difficult to use as a differentiable component in a machine learning pipeline [21].

One way to avoid this repeated optimization is to train a neural surrogate to approximate FBA predictions. The surrogate learns from FBA-generated samples and then predicts fluxes directly from nutrient inputs [61]. Neural networks are good at learning complex relationships, but biological datasets are often too small to train them reliably. Metabolic models can generate structured flux data across many simulated nutrient conditions. Recent studies show that using metabolic knowledge can improve predictions even when experimental data are limited [21, 63].

1.1 Contributions

The main contribution of this thesis is FluxTransformer, a neural surrogate for FBA. It removes the need for a separate optimization stage and instead directly predicts the entire metabolic flux distribution from nutrient uptake bounds. The surrogate uses the Transformer architecture [67], which is now widely used in modeling complex data such as natural language [15, 6] and images [17]. In FluxTransformer, metabolic reactions are represented as tokens with trainable embeddings. Attention allows the model to learn the relationships between reactions without having to specify these dependencies in advance [67]. This is useful since changes in nutrient uptake can affect distant parts of the metabolic network [52].

Earlier studies have combined neural networks with constraint-based metabolic models, for example with Artificial Metabolic Network (AMN) [21] and Metabolic-Informed Neural Network (MINN) [63]. We investigate whether a fully neural Transformer surrogate is able to approximate FBA-generated flux distributions and whether its learned reaction representations contain biologically meaningful structure. The model is evaluated on metabolic networks of increasing size, from a small *Escherichia coli* core model to genome-scale models of *E. coli* and *Saccharomyces cerevisiae*. Further experiments examine whether the surrogate can transfer from simulated flux data to experimental prediction tasks and how training can be made more efficient as the size of the network increases.

1.2 Research Questions

The longer-term goal is to build a single neural surrogate for metabolic prediction across organisms and nutrient conditions, and eventually for microbial communities. To determine whether FluxTransformer provides a feasible starting point, the study asks the following research questions:

RQ1. Can Transformer-based neural surrogates accurately approximate FBA-based metabolic modeling?

RQ2. Does the model capture biologically meaningful relationships between metabolic reactions?

RQ3. How does predictive performance change across metabolic models of increasing size and complexity?

RQ4. How well does the surrogate model transfer from simulated flux data to experimental tasks?

RQ5. How can neural surrogate training be made more efficient for genome-scale metabolic models?

This thesis is organized as follows. Chapter 2 introduces the necessary background on metabolic modeling, Flux Balance Analysis, and neural approaches for metabolic prediction. It also reviews Artificial Metabolic Network [21] and Metabolic-Informed Neural Network [63] as the main related work. Chapter 3 describes the overall approach, FluxTransformer architecture, metabolic models, simulated training data, experimental prediction tasks, and evaluation methods. Chapter 4 presents the results for the *E. coli* core model, *E. coli* iML1515, and *S. cerevisiae* Yeast9, including surrogate performance, experimental prediction tasks, and training-efficiency experiments. Chapter 5 discusses the main findings, limitations, and potential directions for future work. Finally, Chapter 6 concludes the thesis.

2. Background

Predicting microbial behavior means combining two things: a biological description of how metabolism works, and a computational way to turn that description into quantitative predictions. Metabolic models provide the biological side by representing cellular metabolism as a network of reactions and metabolites. The computational side is commonly handled with Flux Balance Analysis, a constraint-based method for predicting metabolic fluxes from these networks [54]. More recently, neural metabolic models have extended this approach. For example, Faure et al. proposed a model that combines a metabolic model with a neural network, so the model respects the underlying biology instead of treating it as a black box [21].

This chapter covers the necessary background for building a neural surrogate for metabolic modeling, including how metabolism is represented in metabolic models in Section 2.1, the FBA method in Section 2.2, and neural approaches to metabolic modeling and related work in Section 2.3.

2.1 Metabolic Modeling

Metabolic modeling is a way to describe how a microorganism uses nutrients to grow, produce energy, and release substances into its environment. These measurable characteristics represent the cellular phenotype. Metabolic models are based on a mathematical description of cellular metabolic processes, where nutrients are converted through connected networks of biochemical reactions [64]. Microorganisms can grow on many raw materials, including industrial side streams and waste compounds, and convert them into biomass, fuels, chemicals, and other valuable products [39]. Metabolic models help researchers simulate how cells will react to varying environmental or genetic conditions. This supports both strain design, where microorganisms are modified to improve production, and bioprocess optimization, where growth conditions are adjusted for better performance [69].

The basic components of a metabolic model are metabolites and reactions. Metabolites are chemical compounds that participate in metabolism: they can be consumed as substrates, formed as products, or passed between reactions as intermedi-

ates. Reactions connect these metabolites into a network by describing the biochemical transformations that occur in the cell [52]. For example, glycolysis converts glucose into pyruvate and produces adenosine triphosphate (ATP) [2]. Stoichiometry defines these transformations quantitatively by specifying how many units of each metabolite are consumed and produced [54].

Reactions are connected into metabolic pathways, where a larger cellular function is performed in several steps. For example, some pathways break down nutrients to release energy, while others use smaller organic molecules to build larger cellular components, such as proteins and nucleic acids [2]. Their activity is described by flux, the rate at which metabolites move through a reaction or pathway [5]. All fluxes together form a flux distribution, which describes one possible metabolic state of the network under specific nutrient conditions [52]. The direction of the reaction is also part of the model structure: irreversible reactions can carry flux only in one direction, while reversible reactions can carry flux in either direction depending on which metabolites are available and how the surrounding network operates [42].

A metabolic model also needs to describe how the cell interacts with its environment. This interaction is described using exchange reactions, which determine which metabolites can enter the metabolic network from the environment or leave it [54, 52]. Uptake refers to nutrients that the organism can consume, such as glucose, oxygen, ammonia, and phosphate. Secretion refers to metabolic products that leave the model, such as acetate, ethanol, lactate, and carbon dioxide. The availability of nutrients is usually set by limiting the uptake rates of the corresponding exchange reactions. These limits define the simulated growth environment and affect which metabolic routes the model can use under those conditions. Realistic input definitions are therefore important to obtain meaningful predictions from constraint-based models [48].

Metabolic models also differ greatly in scale. Small core models describe only a limited part of metabolism, typically central metabolic routes such as glycolysis, the tricarboxylic acid cycle (TCA), and a selected set of exchange reactions. Their smaller size makes them easier to inspect and visualize, which is useful for teaching, exploratory analysis, and method development. The *Escherichia coli* core model is an example of this kind of compact and interpretable representation [53]. Larger genome-scale metabolic models (GEMs) are built from genome-scale reconstructions and aim to represent the known metabolism of an organism as comprehensively as possible. They contain large sets of reactions and metabolites, and many reactions are linked to the genetic basis of the organism through gene–protein–reaction associations [64, 52]. The first GEM was reconstructed for *Haemophilus influenzae* [19], and genome-scale models have since been developed for many other organisms, including *E. coli* and *Saccharomyces cerevisiae* [22, 11]. This thesis uses models of both organisms.

The network structure of metabolic models can be examined visually. Figure 2.1 shows the *E. coli* core metabolic network visualized using `Escher`, a web-based tool for interactive pathway visualization [38]. In this figure, reactions are shown as blue arrows and metabolites as orange nodes. The one-way arrows indicate irreversible reactions, and the two-way arrows indicate reversible reactions. Exchange reactions are at the edges of the map, where metabolites enter or leave the model. The biomass reaction is in the lower right corner and combines several precursor metabolites. Its coefficients show how much each metabolite contributes to forming biomass [53]. Although metabolic pathways are not labeled, recognizable structures can still be identified. For example, the TCA cycle is the ring-shaped structure near the center of the map.

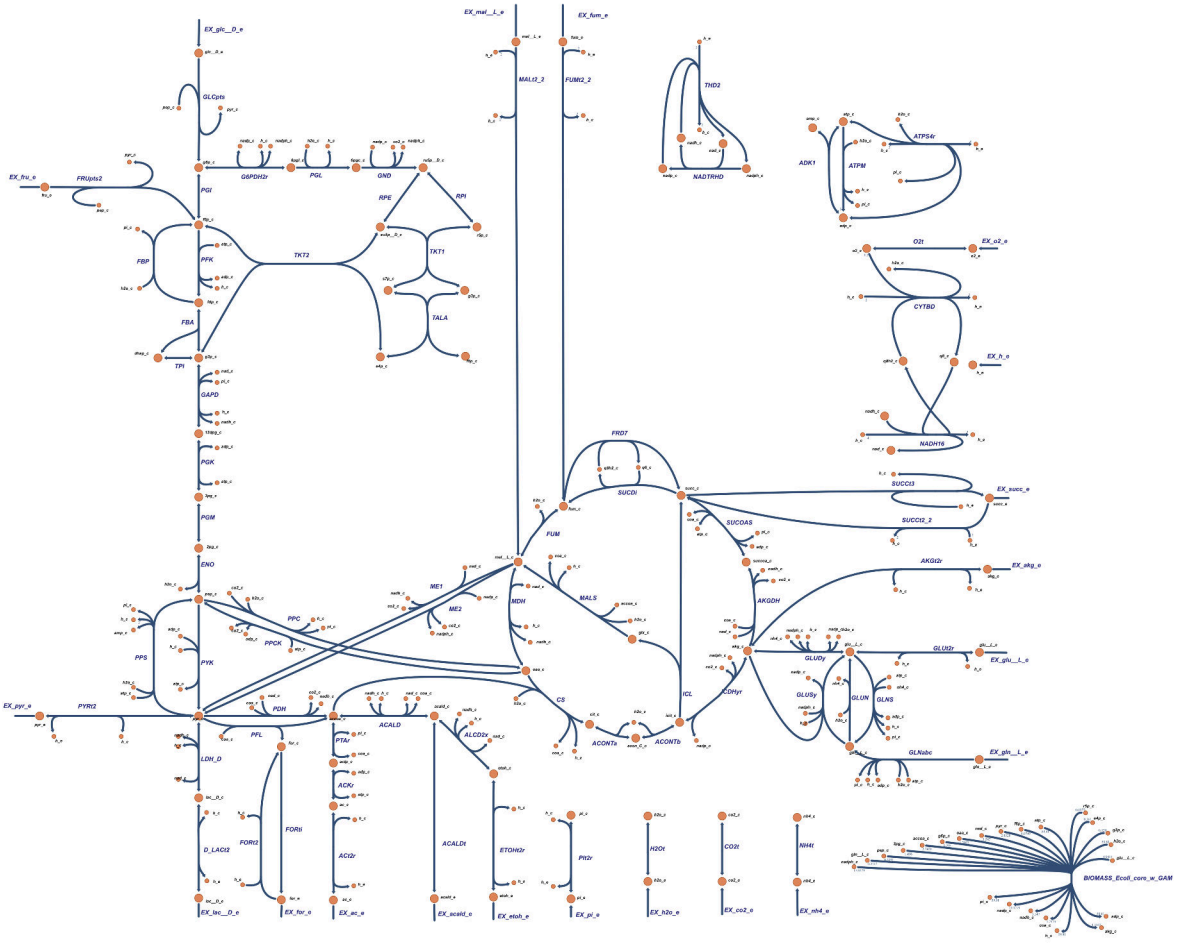


Figure 2.1: Visualization of the *Escherichia coli* core metabolic network generated using the `Escher` pathway visualization tool [38]. Due to the large number of reactions, individual reaction labels are not readable in this image. A zoomable version is available at https://escher.github.io/#/app?map=e_coli_core.Core%20metabolism&tool=Builder&model=e_coli_core.

2.2 Flux Balance Analysis

Flux Balance Analysis (FBA) is a constraint-based modeling approach used to predict flux distributions in metabolic networks [54]. Early work showed that FBA could predict growth and by-product secretion in *E. coli* [66]. FBA assumes that intracellular metabolite concentrations remain constant over time, so that for each metabolite the total production rate equals the total consumption rate. Under this steady-state assumption, cellular metabolism can be represented as a constrained optimization problem. Reaction fluxes are constrained by the stoichiometry of the metabolic network and by lower and upper bounds on individual reactions [57].

FBA also needs an objective function that defines the optimization target. For growth simulations, the model usually assumes that the cell maximizes biomass production [23]. This is done by maximizing the biomass reaction flux, which represents the production of cellular material. The resulting linear optimization problem is then solved using a linear programming method, for example, the Simplex algorithm [37].

The standard formulation of FBA is as follows [54]:

$$\begin{aligned} \max_{\mathbf{v}} \quad & \mathbf{c}^T\mathbf{v} \\ \text{s.t.} \quad & \mathbf{S}\mathbf{v} = 0, \\ & \mathbf{a} \leq \mathbf{v} \leq \mathbf{b}. \end{aligned} \tag{2.1}$$

Here, $\mathbf{v} \in \mathbb{R}^n$ contains the reaction fluxes. The stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m \times n}$ represents m metabolites and n reactions, and $\mathbf{c} \in \mathbb{R}^n$ defines the objective function. The constraint $\mathbf{S}\mathbf{v} = 0$ states that, at steady state, each metabolite is produced and consumed at the same rate. The vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ give the allowed lower and upper limits for each flux. Figure 2.2 illustrates how FBA uses stoichiometric constraints to define feasible flux distributions and select an optimal solution.

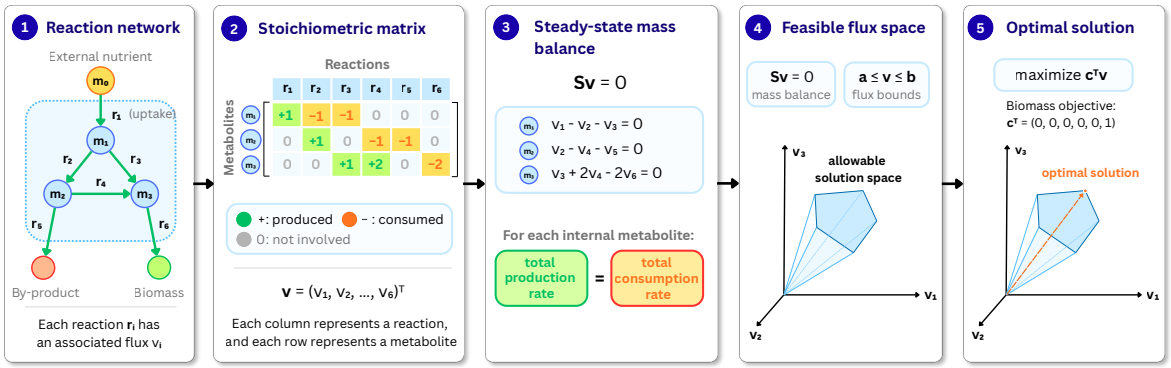


Figure 2.2: Schematic overview of FBA. Stoichiometric constraints and reaction bounds define the feasible flux space, and the objective function selects an optimal solution. Inspired by Orth et al. [54].

Several variants of FBA have been developed for different modeling tasks, including flux variability analysis (FVA), dynamic FBA, regulatory FBA, and parsimonious FBA (pFBA). For an overview of these and other constraint-based methods, see Lewis et al. [42]. Two forms are especially relevant for the models discussed later in this thesis: standard FBA and pFBA. Standard FBA finds a flux distribution that optimizes the chosen objective, but multiple flux distributions can often achieve the same optimum. Parsimonious FBA chooses among these alternatives by selecting the solution with the smallest total flux [41].

For a given nutrient condition, the bounds $\mathbf{a}$ and $\mathbf{b}$ in Eq. 2.1 are fixed and the FBA solver returns an optimal flux distribution $\mathbf{v}$. Changing the nutrient condition changes these bounds and requires solving the optimization problem again. If FBA is embedded in a larger optimization problem, it would be useful to know how changes in the input bounds affect the resulting fluxes. Conventional FBA solvers do not directly provide this gradient information, which limits their use in gradient-based optimization [21].

Another challenge is that condition-specific uptake bounds are rarely available directly from experimental data. They are often assigned manually, and together with the choice of objective function, they can substantially change the resulting flux distribution [48, 60]. FBA also leaves out metabolite concentrations, regulatory effects, and cellular dynamics [5]. These limitations make surrogate models a useful alternative. A learned surrogate can retain the FBA-based input–output structure while being easier to connect with data-driven prediction [61].

2.3 Neural Approaches to Metabolic Modeling

Machine learning has become a useful complement to metabolic modeling because it can learn relationships directly from data [70, 3]. Entirely data-driven models can detect complex patterns, but they commonly require large datasets and can be difficult to interpret biologically. Hybrid models reduce this limitation by combining machine learning with mechanistic information from metabolic models [21, 63].

Training on synthetic or simulated data has become an important strategy when real training data are limited. Tabular foundation models such as TabPFN and TabICL are pretrained on synthetic tabular tasks before being applied to real datasets [29, 58]. Neural operators such as FNO and DeepONet are similarly trained on simulated solutions of differential equations [43, 46]. In this thesis, we follow the same basic principle of these works and train a neural surrogate for metabolic modeling using flux distributions generated with FBA and pFBA. A related use of FBA-generated training data was recently studied by Faure et al. [21], and their model is later used as

a comparison to our model.

Simulated metabolic data have been used in several ways for flux prediction. Neural surrogates can be trained on FBA-generated solutions to replace repeated FBA optimization with faster predictions, for example, in dynamic reactive transport simulations [61]. Can et al. [8] trained several machine-learning models, including an FT-Transformer, to predict FVA-derived flux ranges from metabolomics data. In another data-driven approach, transcriptomic and proteomic data were used to predict the measured fluxes directly, with better performance than pFBA under the conditions tested [25]. FBA-derived reaction-flow graphs have also been used to train graph neural networks to identify genes that are essential for growth [27], and neural networks have been used to infer flux bounds for constraining metabolic models [50].

The following two subsections present the studies most closely related to this thesis: Artificial Metabolic Network (AMN) [21] and Metabolic-Informed Neural Network (MINN) [63].

2.3.1 Artificial Metabolic Network (AMN)

Artificial Metabolic Networks are hybrid neural–mechanistic models designed to make FBA-based predictions compatible with neural network training [21]. They take medium composition C_{med} or uptake-bound information V_{in} as input and predict metabolic fluxes V_{out} or growth rates. The classical FBA optimization step is replaced with differentiable mechanistic layers that approximate the behavior of an FBA solver.

Three mechanistic variants were proposed. AMN-Wt propagates fluxes through the network using learned flux-branching weights. AMN-LP uses a differentiable surrogate of a linear-programming solver. AMN-QP uses a quadratic-programming formulation to refine fluxes. It is inspired by physics-informed neural networks [14] and includes steady-state mass-balance and reaction-bound constraints in its loss function. All AMN variants produce an initial flux estimate using a trainable neural component, which is then refined by the mechanistic component to satisfy metabolic constraints. These hybrid models improved predictions of phenotypes when compared with purely constraint-based modeling and required smaller training sets than pure machine learning methods [21].

The authors also included a fully data-driven neural network as a comparison model, without any mechanistic component. It was trained on the same simulated medium samples as the AMN models, using the full flux distribution as the training target. This model performed worse on the *E. coli* core model and failed to converge on a larger GEM under the reported training setup [21].

The AMN-Reservoir setup is particularly relevant to this thesis. It is a two-

stage framework created for experiments in which condition-specific uptake bounds are unknown, as illustrated in Figure 2.3. In the first step, an AMN is pretrained on FBA-simulated data to learn the mapping from uptake flux bounds to the corresponding flux distribution. It is then frozen and used as a differentiable reservoir that approximates FBA [21]. For the experimental *E. coli* setting, the reservoir was implemented using the AMN-QP variant and trained on a simulated dataset generated from a reduced version of the *E. coli* GEM iML1515 [49].

In the second step, a separate neural preprocessing component is trained on experimental medium compositions to predict uptake flux bounds for the frozen reservoir. The authors refer to this component as an artificial neural network (ANN). In this thesis, it is described more specifically as a multilayer perceptron (MLP). The predicted bounds are passed either to the frozen reservoir or to a standard FBA solver. The overall goal of AMN-Reservoir is to learn the relationship between measurable medium composition and nutrient uptake.

A subsequent study extended the AMN framework to dynamic FBA, where neural–mechanistic models are used to predict time-dependent growth behavior from initial medium conditions [20].

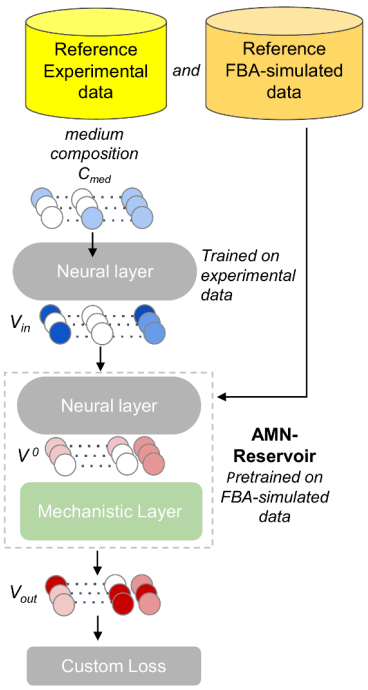


Figure 2.3: Two-stage AMN-Reservoir setup. An AMN is first pretrained on FBA-simulated data and frozen as a reservoir. A neural layer is then trained on experimental medium compositions C_{med} to predict uptake bounds V_{in} for the frozen reservoir. Adapted from Figure 1d in Faure et al. [21].

2.3.2 Metabolic-Informed Neural Network (MINN)

MINN [63] extends AMN to predict metabolic fluxes from transcriptomic, proteomic, and fluxomic data. The model combines a neural network with a mechanistic layer built from a genome-scale metabolic model, allowing omics measurements to guide flux predictions. The authors applied MINN to *E. coli* K-12 MG1655 using reduced versions of the iAF1260 reconstruction [22]. Compared with the iML1515 model used in the AMN study, iAF1260 is an older and smaller reconstruction [22, 49]. Its smaller size reduced the computational resources required to train the hybrid model [63].

In its reservoir-style configuration, MINN is conceptually similar to AMN-Reservoir. The surrogate is first trained on simulated FBA data to predict flux distributions from five variable exchange fluxes: glucose, oxygen, CO_2, ethanol, and acetate. It is then frozen, and an MLP is trained on experimental data to infer exchange constraints from omics measurements and measured fluxes. These constraints are passed to the frozen reservoir and used in pFBA. This two-stage procedure improved the average pFBA-based predictions and reduced variation across the validation splits [63].

2.3.3 Relation to Previous Work

This thesis builds on ideas from the hybrid neural–mechanistic models AMN [21] and MINN [63] for metabolic prediction. The model developed here is a Transformer-based neural surrogate for metabolic flux prediction, referred to as FluxTransformer in this thesis. It is fully neural and trained on FBA-generated flux data. It differs from AMN and MINN in several ways: it has no separate mechanistic layer, retains the original reaction structure, does not split reversible reactions, and uses a Transformer-based representation of metabolic reactions.

The first difference concerns the mechanistic component. Unlike AMN and MINN, FluxTransformer does not include a handcrafted mechanistic layer. Instead, it uses a fully learned Transformer trained on simulated FBA data to learn the relationship between nutrient uptake constraints and steady-state flux distributions. Predictions are produced by a single forward pass through the model.

FluxTransformer also handles the structure of metabolic models differently. Earlier approaches simplified the models before training by removing reactions that were inactive under selected simulation conditions [21, 63]. This work retains the full reaction sets of both *Escherichia coli* iML1515 [49] and *Saccharomyces cerevisiae* Yeast9 [72]. Reaction activity changes with the nutrient condition and the uptake bounds [54]. Keeping the full reaction set avoids a preprocessing bias toward one medium or simulation setup and preserves the original scope of the metabolic model. It also retains reactions that are inactive under those conditions but may become active under other

nutrient conditions represented in the training data.

Reversible reactions are also not separated into individual forward and backward reactions, and exchange reactions are not duplicated for architectural convenience. Instead, FluxTransformer predicts signed fluxes directly, so reaction direction remains part of the learned prediction task. This also preserves a one-to-one correspondence between reaction tokens and reactions in the original metabolic model.

In this thesis, the iML1515 experiments use restricted nutrient settings to allow comparison with AMN and MINN. FluxTransformer was mainly used here to test whether the proposed surrogate concept works on a genome-scale *E. coli* model under comparable conditions. The wider aim of developing a more general model is addressed in the Yeast9 experiments, where the simulated training data cover a more diverse set of nutrient combinations.

FluxTransformer also differs from AMN and MINN in how it represents and connects metabolic reactions. This is described in Section 3.2, where the Transformer architecture and reaction-token representation are introduced.

3. Methods

This chapter presents FluxTransformer and the methods used to train it on simulated metabolic data, apply the pretrained model to experimental prediction tasks, and evaluate its performance.

3.1 Overview of the Approach

FluxTransformer is the neural surrogate developed in this thesis. We first train it on large datasets of simulated metabolic data so that it learns how metabolic fluxes change under different nutrient conditions for a selected metabolic model. The pretrained surrogate is then applied to smaller experimental datasets in downstream prediction tasks. Figure 3.1 summarizes this overall approach.

We generate the pretraining data by varying nutrient uptake bounds in the selected metabolic model and solving FBA or pFBA for each nutrient condition. The uptake bounds and corresponding flux distribution form one training example for FluxTransformer.

Because each organism-specific metabolic model contains a different set of reactions, we train a separate FluxTransformer for each model. We use the *Escherichia coli* core model for architecture development and representation analysis, iML1515 for genome-scale *E. coli* experiments and experimental prediction tasks, and *Saccharomyces cerevisiae* Yeast9 to evaluate the approach on a larger eukaryotic network.

After pretraining, we first evaluate FluxTransformer on unseen simulated nutrient conditions. We then freeze its weights for the experimental tasks. A task-specific neural network is trained to map the available experimental inputs to the uptake bounds used as input by FluxTransformer. In this thesis, this setup is used for growth prediction from medium composition and missing-flux prediction from omics data.

The approach of first pretraining on simulated data and then transferring to experimental tasks is inspired by the AMN-Reservoir described in Section 2.3.1 [21]. Related ideas are also used elsewhere in machine learning. Prior-Data Fitted Networks learn from synthetically generated tasks before being applied to new datasets [51]. TabPFN uses this idea for tabular prediction [29]. Gelman et al. [24] similarly pre-

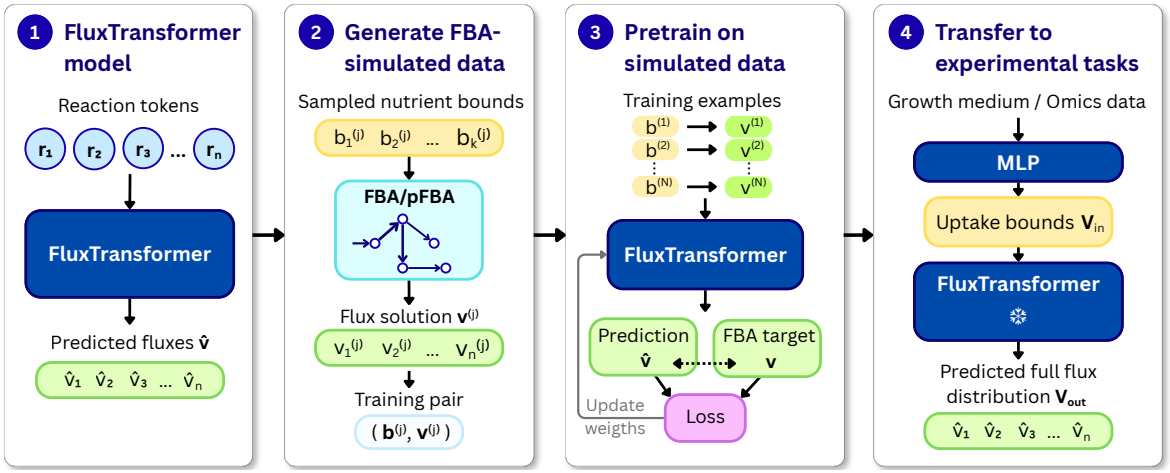


Figure 3.1: Overview of the pretraining of FluxTransformer and its application to experimental tasks.

trained Transformer models on molecular simulations before adapting them to experimental protein data.

Section 3.2 introduces the FluxTransformer architecture. Section 3.3 then presents the metabolic models used in this thesis, followed by the generation of FBA-simulated training data in Section 3.4. The training procedure, experimental datasets and prediction pipelines, and training-efficiency methods are described in Section 3.5. The remaining sections cover comparison models, evaluation, and computational resources.

3.2 FluxTransformer Architecture

The Transformer architecture was originally introduced for machine translation by Vaswani et al. [67]. Its main feature is self-attention, which allows the components of an input sequence to exchange information directly with each other. In language models, these elements are called tokens, which correspond to words, smaller subword units, and other text elements. The same idea can be used with other data types by treating their basic input elements as tokens. For example, in computer vision, an image can be divided into small patches that are treated as tokens [17]. This has allowed Transformer-based models to be applied in many other domains, including audio processing and time-series modeling [44, 68].

In the proposed surrogate, metabolic reactions are represented as tokens. Each token corresponds to one reaction in the metabolic model, and self-attention enables the model to learn interactions between reactions under a given nutrient condition. This is suitable for the prediction of metabolic fluxes because the resulting flux distribution

reflects how the whole metabolic network behaves.

The proposed model is called FluxTransformer. It is trained as a neural surrogate for FBA, and it learns how known flux constraints relate to full flux distributions from FBA simulations. Each training sample contains a set of input flux constraints and the corresponding FBA-predicted flux values for all reactions. The architecture represents each reaction using two coupled quantities: a learned vector embedding X and a scalar flux value c.

In a standard Transformer, each layer updates only the vector representation of each token. FluxTransformer instead updates both the reaction embeddings X and the flux values c together in each layer. Attention controls how strongly reaction tokens interact and how flux information moves between reactions. The model starts with known input constraints and gradually refines the flux values in each layer until it reaches the final flux distribution. We do not use positional encoding because the order of reactions does not matter and has no biological meaning, unlike word order in a sentence.

The parallel c-branch expands the standard Transformer to a setting where an additional scalar value needs to be propagated through the network. Related extensions have been proposed in graph representation learning and protein structure prediction. The Edge-augmented Graph Transformer, for instance, maintains a separate edge channel that evolves together with the node representations and contributes to the attention mechanism [31]. In protein structure prediction, AlphaFold's Evoformer updates two types of representations in parallel and allows information to pass between them throughout the network [35]. To our knowledge, this specific architecture used in FluxTransformer has not been proposed in any previous work.

Figure 3.2 shows a single FluxTransformer layer. The layers can be stacked to form a deeper model. Within each layer, the reaction embeddings X and scalar flux values c are updated using multi-head attention, residual connections, and the feed-forward block.

Embeddings. The model assigns a learnable embedding vector to each reaction token. If n is the number of reaction tokens, and d_{model} is the embedding dimension, the embedding matrix is

$$E \in \mathbb{R}^{n \times d_{\text{model}}}.$$

The rows of E define the reaction embeddings. For reaction token i, the initial embedding is

$$x_i^{(0)} = E_i.$$

The entries of the embedding matrix E are initialized independently as $e_{ij} \sim \mathcal{N}(0, \sigma^2)$, where $\sigma = 1$. During training, these values are updated so that each reaction gets its

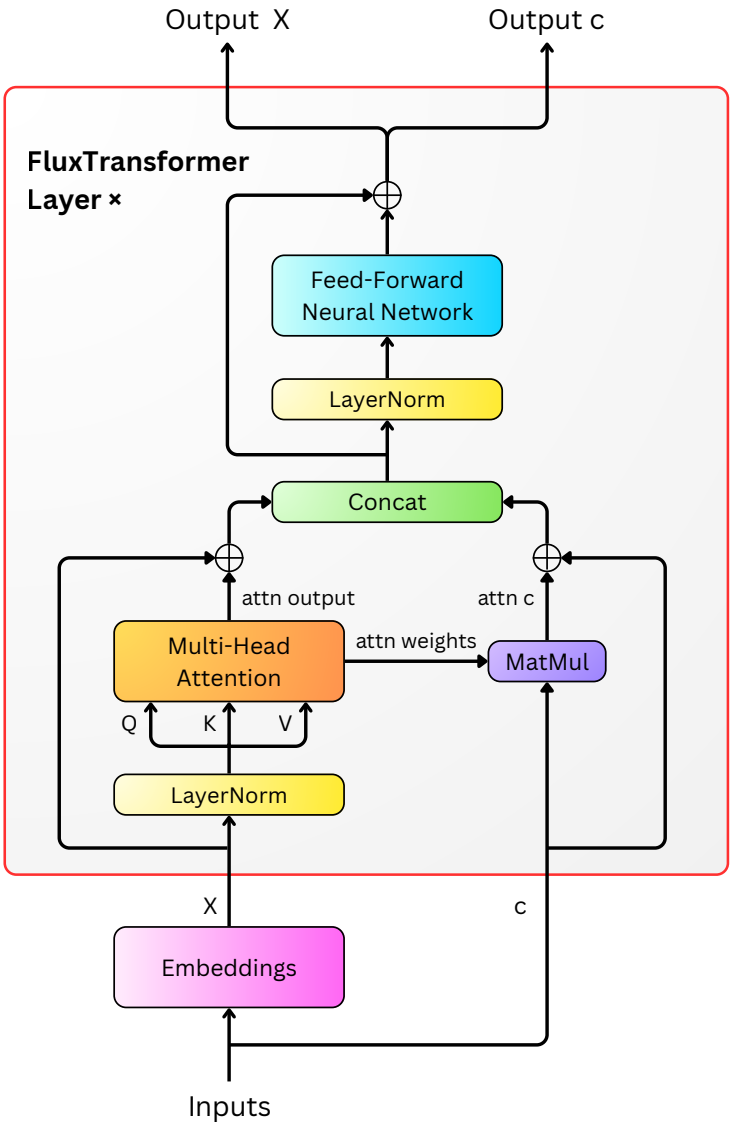


Figure 3.2: Architecture of a single FluxTransformer layer. Reaction embeddings X and scalar flux values c are updated through multi-head attention and a feed-forward block with residual connections. Compared with a standard Transformer layer, FluxTransformer adds a parallel flux branch c that is propagated using the attention weights.

own learned identity, which is shared across all samples. The condition-specific scalar values are stored separately in $c^{(0)}$, allowing them to change with the input condition.

The entries of $c^{(0)}$ are set to the known flux constraints for input reactions, while all other entries are set to zero. If I is the set of input-token indices and v_i^{in} is the known input flux value for reaction token i, this can be written as

$$c_i^{(0)} = \begin{cases} v_i^{\text{in}}, & i \in I, \\ 0, & i \notin I. \end{cases}$$

Architecture. Each FluxTransformer layer jointly updates the reaction tokens and their associated scalar flux values. Let

$$X \in \mathbb{R}^{n \times d_{\text{model}}}, \qquad c \in \mathbb{R}^{n \times 1},$$

where n is the sequence length, equal to the total number of reaction tokens. The batch dimension is omitted from the equations for clarity. Thus, X and c describe one training sample, while the implementation applies the same operations to all samples in a mini-batch.

The token representations are first normalized using layer normalization [4],

$$\widetilde{X} = \text{LayerNorm}(X),$$

and then projected to queries, keys, and values as in standard multi-head attention [67]

$$Q = \widetilde{X}W^Q, \qquad K = \widetilde{X}W^K, \qquad V = \widetilde{X}W^V,$$

where W^Q, W^K, and W^V are learned projection matrices. The queries and keys determine how strongly reaction tokens attend to one another. The values contain the token information passed between reactions, weighted by attention, so that the model learns which reaction representations should influence one another.

To capture different types of interactions in parallel, Q, K, and V are split into H attention heads, each of dimension $d_k = d_{\text{model}}/H$. For attention head h, the attention weights are

$$A_h = \text{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_k}}\right),$$

where Q_h and K_h denote the queries and keys for head h. The resulting matrix $A_h \in \mathbb{R}^{n \times n}$ contains the attention weights between all reaction tokens. Each row shows how much information each reaction contributes when updating one reaction. The corresponding head output is

$$\text{head}_h = A_h V_h.$$

The multi-head attention output is then

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_H)W^O,$$

where W^O is a learned output projection matrix. The residual connection then gives

$$X_{\text{att}} = X + \text{MHA}(Q, K, V).$$

The scalar flux values are updated differently from the token embeddings. The flux values c are mixed across reaction tokens using the attention weights computed from X. For each attention head, this gives

$$c_h = A_h c.$$

The model also learns how much each attention head contributes to this scalar update. Let s_h denote the learnable score of head h, and define the normalized head weights as

$$\alpha_h = \frac{\exp(s_h)}{\sum_{j=1}^{H} \exp(s_j)}.$$

The updated scalar flux values after attention are then

$$c_{\text{att}} = c + \sum_{h=1}^{H} \alpha_h c_h.$$

The residual term c carries the previous flux estimate forward, while the weighted sum adds the new information gathered through attention.

The feed-forward block combines the learned reaction representation with the current flux estimate, so each reaction update depends on both the token state and the scalar flux value. The attention-updated token representations and scalar flux values are first concatenated

$$Z = \text{Concat}(X_{\text{att}}, c_{\text{att}}),$$

and normalized

$$\widetilde{Z} = \text{LayerNorm}(Z).$$

The normalized representation is then passed through a two-layer feed-forward network with a GELU activation function [28] and dropout regularization [62],

$$\text{FFN}(\widetilde{Z}) = W_2 \,\text{Dropout}\Big(\text{GELU}(W_1 \widetilde{Z})\Big).$$

Finally, a residual connection gives

$$Z_{\text{out}} = Z + \text{FFN}(\widetilde{Z}),$$

and the result is split back into updated token representations and scalar flux values,

$$Z_{\text{out}} = [X',\, c'].$$

Stacking several FluxTransformer layers refines both streams. The final flux prediction is taken from the scalar output c' of the last layer, while the final token embeddings can be extracted separately for representation analysis.

3.3 Metabolic Models

This thesis uses three metabolic models for two organisms, *Escherichia coli* and *Saccharomyces cerevisiae*, summarized in Table 3.1. The *E. coli* core model represents central metabolism, while iML1515 and Yeast9 are genome-scale models of *E. coli* and *S. cerevisiae*, respectively.

Table 3.1: Metabolic models used for surrogate model development and evaluation.

Model	Organism	Scope	Reactions	Exchange reactions
E. coli core	*Escherichia coli*	Core	95	20
iML1515	*Escherichia coli*	GEM	2712	331
Yeast9	*Saccharomyces cerevisiae*	GEM	4131	271

Escherichia coli is a common bacterium found in the intestines of humans and other mammals [36]. The laboratory strain *E. coli* K-12 is a well-characterized model organism with extensive genome annotation [59]. Its compact core model contains the central carbon metabolism of *E. coli* and a limited set of exchange reactions [53]. This model was used to develop and test the surrogate architecture and to examine the learned reaction representations. Its smaller scale made it a natural starting point before moving to the larger genome-scale reconstructions.

iML1515 is a genome-scale reconstruction of the metabolic network of *E. coli* K-12 MG1655, including pathways, transport reactions, and exchange reactions that are absent from the core model [49]. This expanded reconstruction was used to evaluate the surrogate model on a substantially larger bacterial model through two experimental tasks: growth prediction from medium composition and missing-flux prediction from omics data.

Saccharomyces cerevisiae, also known as baker's or brewer's yeast, has long been used in baking and fermentation [40]. As a eukaryotic model organism, *S. cerevisiae* has metabolic processes in separate cellular compartments [11]. Yeast9, a consensus genome-scale model of *S. cerevisiae* [72], was included as the largest model considered in this study. It adds both scale and structural complexity to the evaluation.

3.4 Generation of FBA-Simulated Training Data

Figure 3.3 shows the five-step workflow used to generate simulated training data by applying FBA or pFBA to the selected metabolic models. The steps are explained below.

The simulated datasets consisted of nutrient uptake bounds paired with the corresponding flux distributions computed by FBA. For each organism, the SBML model was loaded into `COBRApy` [18], biomass growth was set as the objective, and samples were generated by varying the nutrient composition of the medium. At the same time, a defined set of essential compounds needed for the organism to survive and grow was

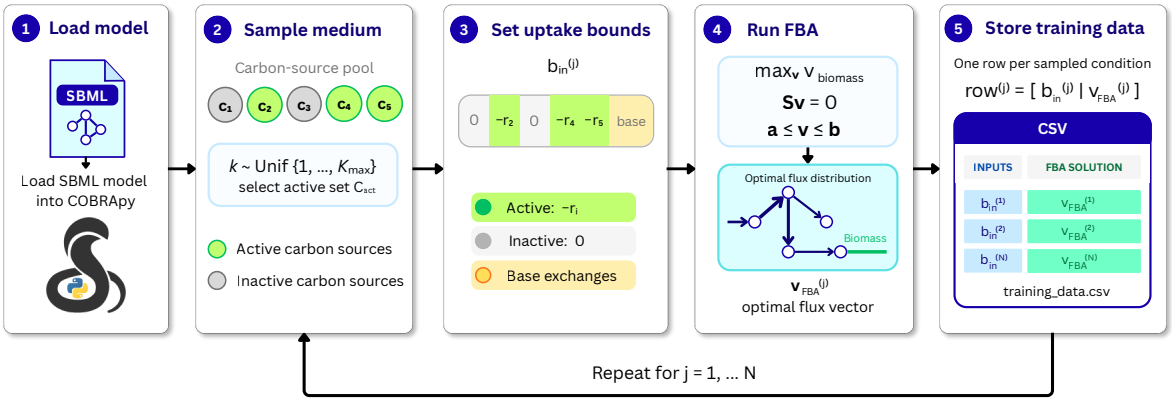


Figure 3.3: Workflow for generating FBA-simulated training data. Each sampled medium is solved with FBA or pFBA, and the input bounds and resulting flux distribution are stored in one row of the training dataset.

kept available in all simulations.

The choice of simulated training conditions directly affects how well the surrogate performs on later data. The simulated data must at least include the nutrient types and uptake ranges that may appear in the test data. A more diverse set of conditions makes the pretrained model more general and useful for a wider range of downstream tasks. The drawback is that, with a fixed number of simulated samples, each part of this larger condition space is represented by fewer examples. As the range of nutrient conditions increases, more simulated samples are therefore needed to keep the training data sufficiently dense.

Each simulated medium is defined by variable carbon sources and a set of base medium components. The carbon-source inputs are first sampled by selecting a random subset from the model-specific carbon source pool $\mathcal{C}$. The number of selected carbon sources, k, is chosen between one and a specified maximum $K_{\max}$, and then that many distinct carbon sources are sampled from the full list to form the active set $\mathcal{C}_{\text{act}}$. Each carbon source i is then assigned an input lower bound b_i^{in} for its exchange reaction. This bound determines its maximum uptake rate, so the samples vary both in which carbons are present and in how strongly they can be consumed.

This procedure can be written as

$$k \sim \text{Unif}\{1, \ldots, K_{\max}\},$$

$$\mathcal{C}_{\text{act}} \sim \text{Unif}\Big(\{A \subseteq \mathcal{C} : |A| = k\}\Big),$$

and

$$b_i^{\text{in}} = \begin{cases} -r_i, & i \in \mathcal{C}_{\text{act}}, \quad r_i \sim \mathcal{D}_C(a_C, b_C), \\ 0, & i \notin \mathcal{C}_{\text{act}}. \end{cases}$$

For each active carbon source i, the value r_i is the sampled uptake limit. The parameters a_C and b_C define its sampling range. The negative sign follows the usual FBA convention, where uptake is represented by negative flux [48]. For carbon sources outside $\mathcal{C}_{\text{act}}$, the bound is set to zero to prevent uptake. The notation $\mathcal{D}_C(a_C, b_C)$ denotes the distribution used to draw uptake limits. Depending on the sampling setup, this distribution was uniform or log-uniform over (a_C, b_C). Log-uniform sampling places more samples in the lower part of the range on the original scale. This gives more detail for small uptake values, while larger values are still included.

After the sampled carbon sources are defined, base exchanges representing the rest of the medium are added, such as oxygen, ammonium, phosphate, water, ions, and trace metals. Most of these nutrients are usually kept at fixed bounds, whereas oxygen and ammonium can either be fixed or treated as variable inputs depending on the complexity desired for the training setup. FBA is then run for each sampled medium by maximizing the biomass objective. The sampled exchange bounds and resulting optimal fluxes are stored in a single training-data row, with the bounds first, followed by the fluxes.

Model-specific input designs. Table 3.2 summarizes the model-specific input designs and dataset sizes. The exact set of inputs for each model varied according to the training configuration. The *E. coli* core model used all 20 exchange reactions as input. This setup included 12 variable carbon sources and additional variable non-carbon exchanges, including oxygen and separately sampled nitrogen sources.

Two different input designs were used for iML1515. The first setup ("iML1515 growth" in Table 3.2) had a reduced subset of nutritionally relevant exchanges as input. The subset was defined to match the experimental setting of AMN-Reservoir, giving 38 inputs [21]. In this setup, ten carbon sources were treated as variable inputs, while glycerol, four amino acids, and the remaining base nutrients were kept fixed. The exception to the AMN-Reservoir setup was oxygen, which was kept fixed in AMN but treated here as a variable exchange. This allowed more variation in aerobic conditions.

The other iML1515 input setup was used for the experiment of predicting missing fluxes, shown as "iML1515 missing flux" in Table 3.2. This setup followed the MINN-Reservoir setting [63], where uptake bounds for glucose and oxygen and secretion bounds for carbon dioxide, ethanol, and acetate were sampled independently. The remaining base nutrients were kept fixed, giving 27 total inputs. As in MINN-Reservoir, these samples were generated using pFBA.

Finally, for Yeast9, the training setup had 64 inputs, including variable carbon sources, essential base nutrients, and variable oxygen and ammonium exchanges. Unlike iML1515, the *E. coli* core and Yeast9 setups were designed for more general

sampling over nutrient conditions. The restricted iML1515 setups were used primarily to benchmark against experimental data.

Dataset sizes. The *E. coli* core dataset contained 10 million samples, both iML1515 datasets contained 500,000 samples, and the Yeast9 dataset contained 250,000 samples. During model development, smaller datasets were used to allow faster iteration while the architecture and training setup were still being adjusted. Preliminary experiments showed that prediction accuracy continued to improve as more simulated samples were added. The final dataset sizes were therefore chosen as a practical balance between prediction accuracy and training time, since training the genome-scale models with these datasets already required several days. The smaller core model allowed a substantially larger dataset to be used at manageable computational cost.

Table 3.2: Model-specific input designs and FBA-simulated dataset sizes.

Model	Var. carbon	Other variable	Fixed inputs	Total	Samples
E. coli core	12	4	4	20	10M
iML1515 growth	10	1	27	38	500k
iML1515 missing flux	1	4	22	27	500k
Yeast9	49	2	13	64	250k

3.5 Training and Experimental Setup

3.5.1 Training on Simulated Data

FluxTransformer was trained on FBA-simulated datasets described in Section 3.4. For each sample, the known exchange flux constraints defining the nutrient condition were placed in the scalar vector c at the corresponding reaction-token positions. The FBA-computed flux distribution was used as the target.

Each simulated dataset was divided into training and validation sets using a fixed 80/20 split. A fixed random seed was used to ensure consistent comparisons between model configurations.

The main model hyperparameters are $\{d_{\text{model}}, H, L, d_{\text{ff}}, p_{\text{drop}}\}$, where d_{model} is the embedding dimension, H is the number of attention heads, L is the number of Transformer layers, d_{ff} is the feed-forward dimension, and p_{drop} is the dropout probability. Selecting these hyperparameters requires repeated training with alternative configurations, which is computationally costly for genome-scale models. In this thesis, most of the architecture selection was therefore carried out with the smaller *E. coli* core model,

where more configurations could be evaluated within the available computational budget. Twelve manually selected architectures were tested, with the candidates chosen to cover meaningful combinations of the hyperparameters. Their performance metrics and biomass-flux predictions are reported in Appendix A.

For iML1515 and Yeast9, only a smaller set of architectures was evaluated because each training run took considerably longer. Automatic hyperparameter optimization methods, such as Optuna [1], could also be used, but would typically require evaluating more candidate configurations.

Table 3.3 summarizes the selected FluxTransformer configurations, mini-batch size, and number of trainable parameters for each metabolic model. The mini-batch size was chosen to fit within the available GPU memory.

Table 3.3: FluxTransformer configurations for each metabolic model.

Model	d_{model}	H	L	d_{ff}	p_{drop}	Batch size	Params. (M)
E. coli core	128	8	4	1024	0.02	128	1.340
iML1515	256	8	3	1024	0.02	12	3.070
Yeast9	256	8	3	1024	0.02	8	3.433

Training was performed in mini-batches by minimizing the mean Huber loss [30]. The Huber loss behaves like squared error for small residuals but becomes linear for larger ones, making it less sensitive to large errors than mean squared error. This was useful because flux magnitudes can vary substantially between reactions. The parameters were optimized using AdamW [45] with a learning rate of 10^{-4}, beta values $(0.9, 0.98)$, $\epsilon = 10^{-9}$ and weight decay 10^{-4}. Gradient norms were clipped to a maximum value of 1.0 after backpropagation. Validation loss was monitored during training, and the checkpoint with the lowest validation loss was saved for testing. Final predictive performance was evaluated on a separately generated test set created with the same sampling procedure but a different random seed.

After training, FluxTransformer was frozen and used as a differentiable surrogate in two experimental prediction tasks. In each task, a neural network was trained to map the experimental data to the input constraints required by the surrogate. The following subsections describe the growth-prediction and missing-flux experiments.

3.5.2 Growth Prediction from Medium Composition

Experimental growth data. The experimental growth dataset was obtained from the GitHub repository accompanying the AMN article by Faure et al. [21, 7]. It contains 110 growth conditions for *E. coli* strain DH5-α. Cells were grown in an M9-based

medium supplemented with four amino acids and ten variable carbon sources. Each of the ten carbon sources was encoded as a separate binary input variable indicating whether that compound was present or absent in the medium. The 110 conditions consisted of all 10 single-carbon media, together with 20 two-carbon, 40 three-carbon, and 40 four-carbon combinations. For each medium condition, optical density was measured over time in 2–8 technical replicates. A maximum growth rate was estimated separately from each replicate curve. The training target was the mean maximum growth rate for that condition, and the standard deviation described the variability between replicates [21].

Prediction setup. The growth-prediction experiment followed the AMN growth-prediction setting [21], with the AMN model replaced by the FluxTransformer surrogate. The experimental inputs show which carbon sources were present in each medium, while the FluxTransformer requires uptake bounds as input. Therefore, the uptake bounds for the variable carbon sources had to be inferred indirectly from the measured growth rates.

The experiment used the FluxTransformer trained with the "iML1515 growth" configuration described in Section 3.4. Its parameters were kept fixed throughout this task. A small multilayer perceptron (MLP) was placed before the surrogate, as shown in Figure 3.4A. Its role was to map the binary medium-composition vector C_{med} to uptake bounds for the variable exchange reactions. Medium components that were constant across all experimental conditions were inserted as fixed uptake bounds. The predicted variable bounds and the fixed bounds together formed V_{in}, the uptake-bound vector passed to the FluxTransformer.

The MLP had one hidden layer with 512 hidden units, ReLU activation, and no dropout. Its output was computed as

$$f_{\text{MLP}}(C_{\text{med}}) = W_2\text{ReLU}(W_1 C_{\text{med}} + b_1) + b_2,$$

where W_1 and W_2 are weight matrices and b_1 and b_2 are bias vectors. The MLP output was multiplied element-wise by C_{med}, so the predicted uptake bound was set to zero for carbon sources absent from the medium.

The FluxTransformer used V_{in} to produce V_{out}, the predicted steady-state flux distribution. Its biomass reaction was compared with the measured mean maximum growth rate, and resulting gradients were passed through the FluxTransformer to update the MLP. The learned uptake bounds are latent variables: the model infers them from growth data to improve prediction, since the experiment itself does not provide measured uptake rates.

Model performance was evaluated using stratified 10-fold cross-validation. The

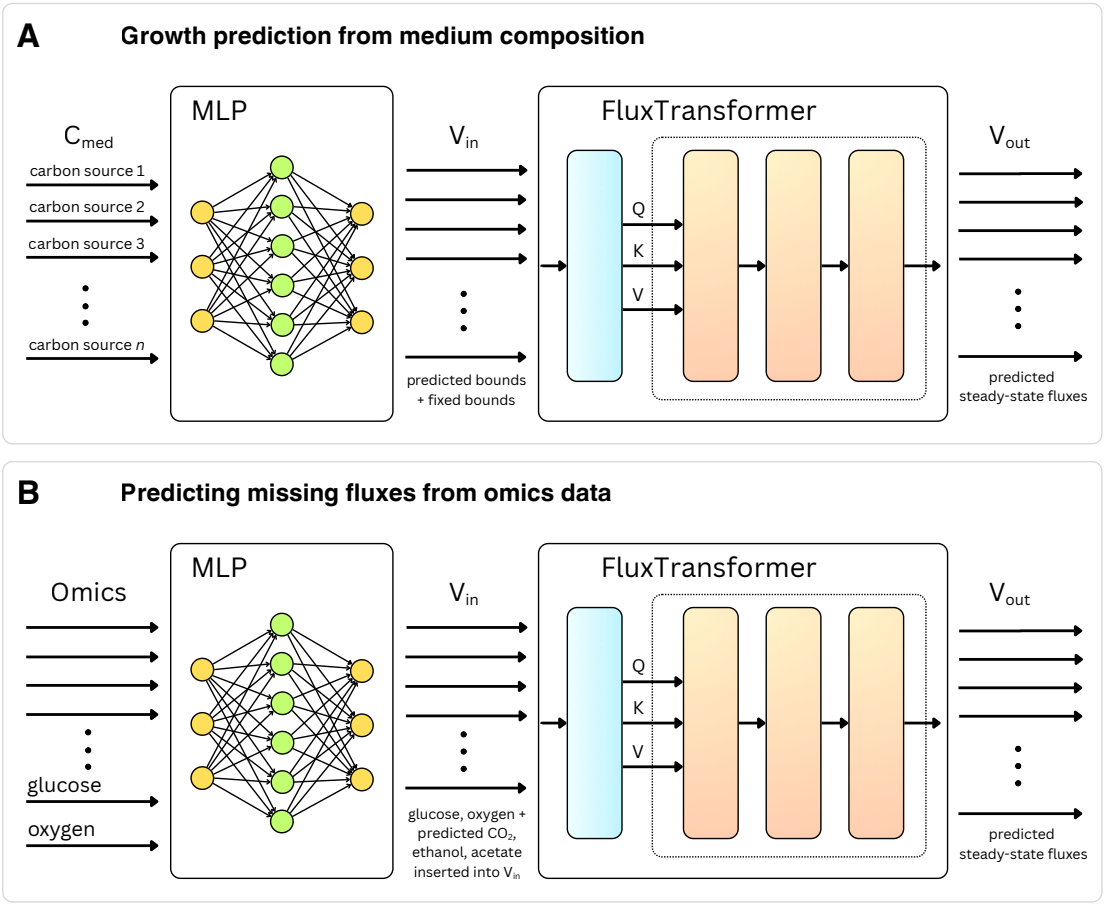


Figure 3.4: Experimental prediction pipelines using a pretrained FluxTransformer. **(A)** In growth prediction, the medium-composition vector C_{med} is mapped by a multilayer perceptron (MLP) to V_{in}, the uptake-bound input vector for the FluxTransformer. The model predicts V_{out}, and the biomass reaction in this flux vector is compared with measured growth. **(B)** In missing-flux prediction, transcriptomic and proteomic measurements are combined with glucose and oxygen uptake. The MLP predicts missing exchange constraints for CO_2, ethanol, and acetate, which are used as inputs to the frozen FluxTransformer and downstream pFBA evaluation.

folds were stratified by the number of variable carbon sources in the medium, with samples grouped into media containing 1, 2, 3, or 4 variable carbon sources. This kept the distribution of these groups similar across folds. Cross-validation was repeated three times using different split seeds.

In each fold, the MLP was trained with a fixed training seed and early stopping based on validation loss. Predictions for the held-out samples were collected across all ten folds. Performance was calculated from the combined held-out predictions and summarized across the three cross-validation repeats using the mean and standard deviation of R^2 and MAE.

3.5.3 Predicting Missing Fluxes from Omics Data

Experimental omics data. The missing-flux prediction experiment used the *E. coli* K-12 dataset originally reported by Ishii et al. [33]. The dataset measures how *E. coli* metabolism changes across several growth rates and after single-gene deletions. The same dataset was later used to evaluate MINN [63]. The data consist of 29 chemostat samples grown in a glucose minimal medium. Five samples come from the unmodified *E. coli* strain grown at dilution rates of 0.1, 0.2, 0.4, 0.5, and 0.7 h^{-1}, where 0.2 h^{-1} is the reference condition. The other 24 samples are single-gene knockout strains, all grown at the same reference dilution rate. For each sample, the dataset includes transcriptomic data for 79 genes, proteomic data for 60 proteins, and measured values for 47 metabolic fluxes [33, 63].

The original Ishii fluxomics data were reported to lie outside the FBA solution space of the metabolic model used in MINN. Because of this, an FBA-fitted version of the flux data was used in the MINN experiments [63]. This thesis uses the same fitted fluxomics file for its main evaluation, allowing direct comparison with the reported pFBA and MINN-reservoir+pFBA benchmarks.

A separate MINN-like fitting was also performed for iML1515. Each sample was adjusted so the flux vector would be feasible for iML1515, with minimal changes to the values. The biomass flux stayed fixed at its original value, and the fitting process minimized the total change in the other fluxes. This fitted version was used for an additional iML1515-specific evaluation.

Prediction setup. The missing-flux experiment used a similar method to the growth-prediction task. The frozen surrogate was the FluxTransformer trained with the "iML1515 missing flux" configuration described in Section 3.4. A small trainable MLP was placed before it, as shown in Figure 3.4B. Here, the MLP's role was different: instead of mapping medium composition to uptake bounds, it inferred missing exchange constraints from omics measurements and two measured fluxes.

The model input combined transcriptomics, proteomics, and the glucose and oxygen uptake fluxes. Glucose and oxygen were used as available exchange constraints, while carbon dioxide, ethanol, and acetate were treated as missing exchange constraints.

The experiment was adapted from the MINN-Reservoir comparison in Table 4 of Tazza et al. [63]. In that setting, predicted exchange constraints are added before running pFBA. Here, the MLP inferred the three missing constraints from omics data and glucose and oxygen uptake fluxes. The predicted values were combined with the glucose and oxygen constraints and the fixed base-medium bounds to form V_{in}, which

was passed to the pretrained FluxTransformer.

The FluxTransformer produced a predicted steady-state flux distribution, which was evaluated against the 47 target fluxes available for each sample. In a separate pFBA evaluation, the predicted missing exchange bounds were used as extra constraints.

Only the MLP parameters were trained in this experiment, while the FluxTransformer parameters stayed fixed. As in the growth-prediction task, the inferred fluxes act as latent variables. The model estimates them from indirect data rather than direct measurements.

Model performance was evaluated using leave-one-out cross-validation (LOO-CV). In each split, the front MLP was trained on 28 samples and evaluated on the remaining held-out sample. Repeating this procedure for all 29 samples produced one held-out predicted flux vector for each sample. The resulting 29 predictions were evaluated against the corresponding target fluxes.

3.5.4 Training-Efficiency Methods for Genome-Scale Models

Each metabolic reaction requires its own token, so the sequence length n grows with the number of reactions in the metabolic model. The computational cost of self-attention scales quadratically with sequence length, as $O(n^2)$ [67]. This makes training increasingly expensive for genome-scale models. The *Saccharomyces cerevisiae* Yeast9 model contains 4,131 reactions and was the largest metabolic model used in this work. We examined this scaling issue using output-subset training on Yeast9. Several subset-sampling variants were tested to compare different ways of choosing the output reactions.

The baseline model processed all input and output reaction tokens in each batch and produced a complete flux vector. Loss was evaluated over every output reaction, giving direct supervision but increasing the cost of each update as the number of reaction tokens grew.

With output-subset training, the model still received all input tokens in each batch. The nutrient uptake bounds therefore remained available during the forward pass. The sequence was shortened only on the output side: FluxTransformer processed the input context together with a sampled set of output reactions, and the loss was evaluated for that sampled set. Let M denote the total number of output reactions and $\mathcal{S}_b$ the subset selected for batch b. The output-sampling ratio p determined the size of this set:

$$|\mathcal{S}_b| = \max(1, \lfloor pM \rfloor) .$$

The evaluated ratios were $p \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, with $p = 1$ representing full-

output training.

All sampling variants used the same model architecture and only changed how the output subsets were chosen. Uniform sampling simply selected output reactions with equal probability, giving each reaction an inclusion probability of $|\mathcal{S}_b|/M$. The other variants used information from the training data to guide the selection. Weighted sampling favored reactions with larger flux activity by assigning reaction i a weight based on $w_i = \log(1 + \overline{|v_i|})$, where $\overline{|v_i|}$ is its mean absolute flux. Correlation-based sampling built subsets from reactions whose fluxes varied together, using the pairwise correlations $C_{ij} = \mathrm{corr}(v_i, v_j)$. Sample-driven sampling first selected a training sample s at random and defined its active set as $\mathcal{A}_s = \{i : v_{s,i} \neq 0\}$. Reactions were sampled from this set, with zero-flux reactions used to fill any remaining positions in the output subset. Weighted, correlation-based, and sample-driven sampling also included random draws during subset construction, so the selected subsets were not limited to just high-flux, correlated, or active reactions.

3.6 Comparison Models and Baselines

The experimental pipelines were compared with task-specific published models and baselines. For growth prediction, performance was compared with the reported AMN-QP, AMN-LP, and AMN-Wt results from Faure et al., which used the same experimental growth dataset and repeated stratified 10-fold cross-validation [21].

An alternative to a dedicated neural metabolic surrogate is to train a black-box machine learning model directly for the experimental prediction task. Such a model does not approximate metabolism or predict a complete flux distribution, but it may still be sufficient if the target can be predicted accurately from the experimental inputs alone. To investigate how well black-box models work, TabPFN-3 was included as an additional fully data-driven baseline using the same binary medium-composition features. TabPFN has shown strong performance in recent tabular data benchmarks and requires minimal task-specific tuning [29, 26].

TabPFN-3 is pretrained on synthetic tabular datasets and uses the labeled training samples as context when predicting unseen samples. It can be applied to a new prediction task without separate model training. In this comparison, TabPFN-3 received only the binary medium-composition features and had no access to FBA-simulated data or the metabolic model. Its performance tests whether metabolic pretraining improves prediction beyond what can be learned directly from the experimental data.

For missing-flux prediction, performance was compared with the pFBA and MINN-Reservoir + pFBA results reported in Table 4 of Tazza et al. [63]. These results used the FBA-fitted fluxomics data described in Section 3.5.3. Since FluxTransformer

used iML1515, a corresponding pFBA baseline was also run with iML1515 using glucose and oxygen as constraints. For the FluxTransformer + pFBA comparison, the predicted carbon dioxide, ethanol, and acetate constraints were added to this same pFBA setup. The final comparison used the fluxomics data fitted specifically to the iML1515 solution space.

3.7 Evaluation Methods

3.7.1 Prediction Metrics

Predictive performance was evaluated using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE). These metrics were selected to allow comparison with previous work: R^2 was used in the AMN study [21], while R^2, MAE, and RMSE were reported for MINN [63]. Training instead used Huber loss because it is less sensitive to large errors. For true values y_i, predicted values $\hat{y}_i$, sample mean $\bar{y}$, and n samples, the evaluation metrics are defined as

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}, \tag{3.1}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{3.2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{3.3}$$

For the task of predicting missing fluxes from omics data, normalized error (NE) was also used. NE measures the prediction error relative to the scale of the flux vector, which makes it easier to compare errors across fluxes with different magnitudes. It was computed as

$$\text{NE} = \frac{\|\hat{\mathbf{y}} - \mathbf{y}\|_2}{\|\hat{\mathbf{y}}\|_2}, \tag{3.4}$$

where $\mathbf{y}$ is the vector of true flux values and $\hat{\mathbf{y}}$ is the vector of predicted flux values.

3.7.2 Representation Analysis

The representation analysis examined the reaction embeddings learned by FluxTransformer. The trained *E. coli* core model was evaluated on sampled nutrient conditions, and the corresponding reaction embeddings were collected from the FluxTransformer

output. Each embedding shows how the model represents a reaction under one nutrient condition.

The embeddings originally have 128 dimensions, which were reduced to two dimensions using t-SNE [65]. t-SNE places similar embeddings close together in the resulting projection, making cluster structure easier to see. The primary visualization included all 95 reaction tokens across 1000 sampled nutrient conditions, giving one point per reaction per condition.

The t-SNE projections were colored either by individual reaction identity or by broader reaction groups, such as metabolic pathways. Selected reactions were also visualized separately to examine specific regions of the representation space in more detail. For these subset plots, nutrient conditions were ordered by Euclidean distance in normalized input space. The conditions were colored from purple to red, so the most divergent conditions appeared at opposite ends of the spectrum.

This analysis examined whether the model learned biologically meaningful relationships between reactions. The visualizations provide exploratory evidence of structure in the learned embedding space, while flux prediction is the main performance measure.

3.8 Computational Resources

Model training and evaluation were performed using two computing environments. The experiments with the *E. coli* core model were carried out on a Lenovo Legion Pro laptop equipped with an AMD Ryzen 7 5800H CPU with 16 GB of RAM, and an NVIDIA GeForce RTX 3070 GPU with 8 GB of memory. Training for the genome-scale models was performed on the Puhti supercomputing platform provided by CSC – IT Center for Science, Finland [12]. The training nodes used Intel Xeon Gold 6230 CPUs and NVIDIA Volta V100 GPUs with 32 GB of memory. The laptop was used for test-set predictions, experimental prediction tasks, evaluation metrics, and post-training analyses and visualizations.

All code specific to this thesis, including the FluxTransformer architecture, FBA simulation pipeline, model training procedures, and evaluation code, was implemented by the author in `Python` 3.12. The main software libraries used were `PyTorch` [55] for neural network modeling and training, `COBRApy` [18] for metabolic modeling and Flux Balance Analysis, and `scikit-learn` [56] for evaluation and data analysis tasks.

4. Results

This chapter presents the FluxTransformer results across simulated and experimental prediction tasks. The first results concern the *E. coli* core model, which was used to test the core surrogate task: predicting FBA-computed fluxes from simulated nutrient conditions. Reaction embeddings from the same model were analyzed to determine whether the learned representations show meaningful structure. The iML1515 experiments evaluate the surrogate on a genome-scale *E. coli* model to assess scalability and transfer from simulated training data to experimental growth and missing-flux prediction. The final section evaluates flux-prediction performance on the larger eukaryotic Yeast9 model of *S. cerevisiae* and compares output-subset training strategies.

4.1 *Escherichia coli* Core Model

4.1.1 Flux Predictions

The *E. coli* core model was the first benchmark because its small size allowed more efficient evaluation than the genome-scale models. Each training sample corresponds to a simulated growth condition defined by the model input constraints, and the surrogate predicts the corresponding FBA-computed flux distribution.

Figure 4.1 shows that the FluxTransformer model trained on 10 million simulated samples predicts biomass flux with high accuracy. Biomass is a useful first diagnostic because it summarizes the combined effect of nutrient availability and intracellular flux allocation on growth. The close agreement with FBA indicates that the surrogate captures how changes in the input constraints affect biomass production.

In the left panel, the predicted values follow the identity line. Biomass fluxes in the test set range from near zero to approximately 2.5. Compared with this range, the MAE of 0.009 and RMSE of 0.012 are small. The R^2 is 0.999. In the right panel, the residuals are centered near zero and show no clear systematic trend across the biomass range. The model is therefore equally accurate at low and high biomass fluxes.

The model was then evaluated across all 95 fluxes in the *E. coli* core network. The pooled predictions had an overall R^2 of 0.991, with an MAE of 0.304 and an RMSE of

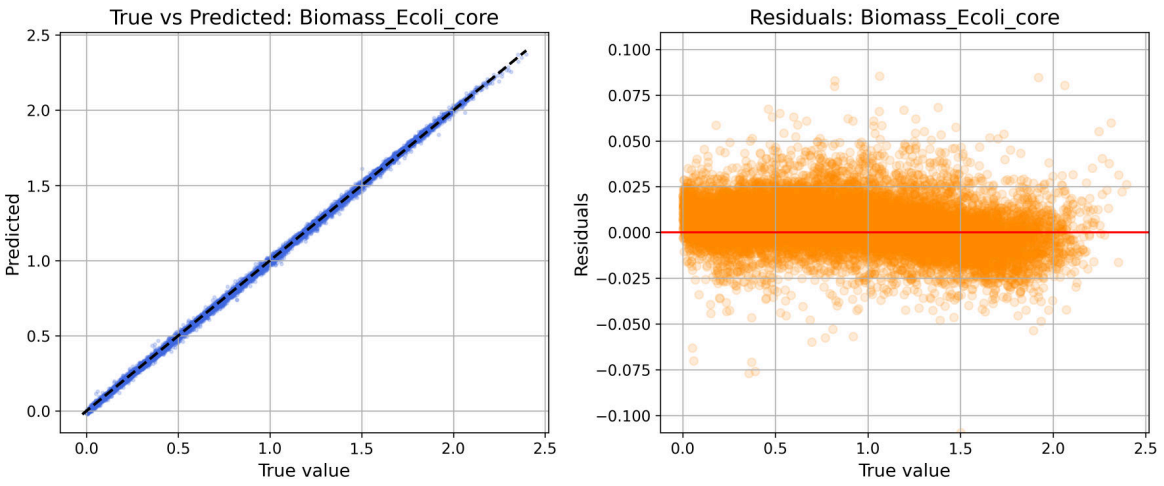


Figure 4.1: Biomass flux prediction for the *E. coli* core model. **Left:** Predicted versus true biomass flux. **Right:** Residuals against the true biomass flux, where each residual is calculated as predicted minus true flux. The model was trained on 10 million simulated samples.

1.122. These metrics indicate close agreement between the predicted and FBA fluxes across the reaction network. At the reaction level, 78 of 95 fluxes had $R^2 \geq 0.90$, 68 had $R^2 \geq 0.95$, and 37 had $R^2 \geq 0.99$. Full per-flux metrics are reported in Appendix B. Reactions with lower R^2 values often had target distributions dominated by values near zero, a strong imbalance between active and inactive samples, or few high-flux observations. Figures 4.2 and 4.3 show two examples: GLUDy, a glutamate dehydrogenase reaction involved in ammonia assimilation, and GLUN, a glutaminase reaction that converts glutamine to glutamate [53].

Figure 4.2 shows the GLUDy predictions in the left panel and the distribution of true fluxes in the right panel. The true flux distribution includes many zero or low-flux samples and spans both negative and positive ranges. A small proportion of samples have large positive fluxes. The scatter plot forms a V-shaped pattern, with one trend close to the one-to-one line and another with systematically lower predictions for similar true fluxes. The predicted range is narrower than the true range, and the largest positive GLUDy fluxes are underpredicted. The model therefore predicts the branch near the identity line more accurately than the lower branch. The plot alone does not explain why the samples form two prediction branches.

For GLUN, the true-flux distribution in Figure 4.3 is dominated by a large peak at zero because the reaction is inactive in most samples. When GLUN is active, many predictions are close to the identity line. The largest errors occur when the true flux is zero, as the model often predicts a positive value even though the FBA solution keeps GLUN inactive. The dominant error is therefore false activation of the reaction.

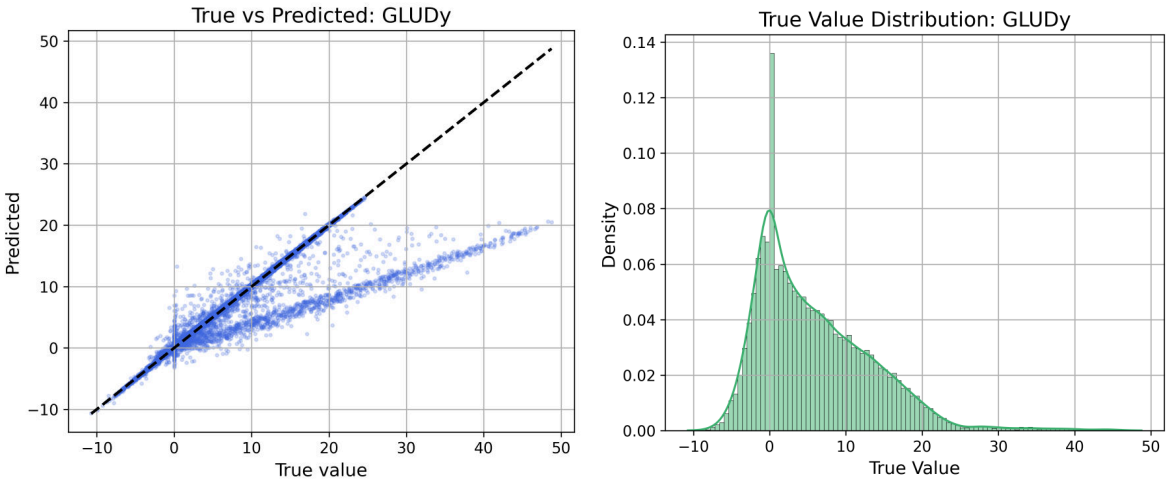


Figure 4.2: GLUDy flux prediction for the *E. coli* core model. **Left:** Predicted GLUDy flux plotted against the true flux. **Right:** Distribution of true GLUDy flux values in the test set.

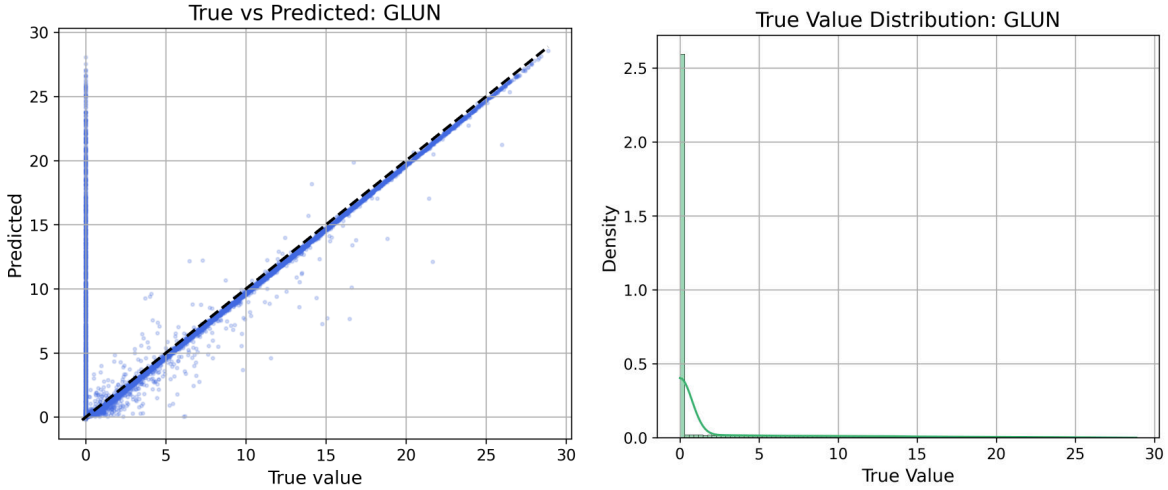


Figure 4.3: GLUN flux prediction for the *E. coli* core model. **Left:** Predicted GLUN flux plotted against the true flux. **Right:** Distribution of true GLUN flux values in the test set.

4.1.2 Learned Reaction Embeddings

To examine how FluxTransformer represented different reactions, the final layer embeddings from the *E. coli* core model were projected to two dimensions using t-SNE. The resulting projection was examined for reaction-specific clusters and relationships among reactions.

Figure 4.4 shows the embeddings from 1000 sampled nutrient compositions. Each point represents one reaction under one sampled condition, and colors distinguish the 95 reactions in the core model, giving 1000 embeddings per reaction. Most reactions occupy distinct regions of the projected space, indicating that reaction identity remains recognizable across varying nutrient inputs. The shapes of these regions also differ. Some clusters are relatively compact, whereas others vary more in shape or divide into nearby subclusters. Because t-SNE does not preserve global distances, the apparent cluster size and spacing should be interpreted qualitatively [65].

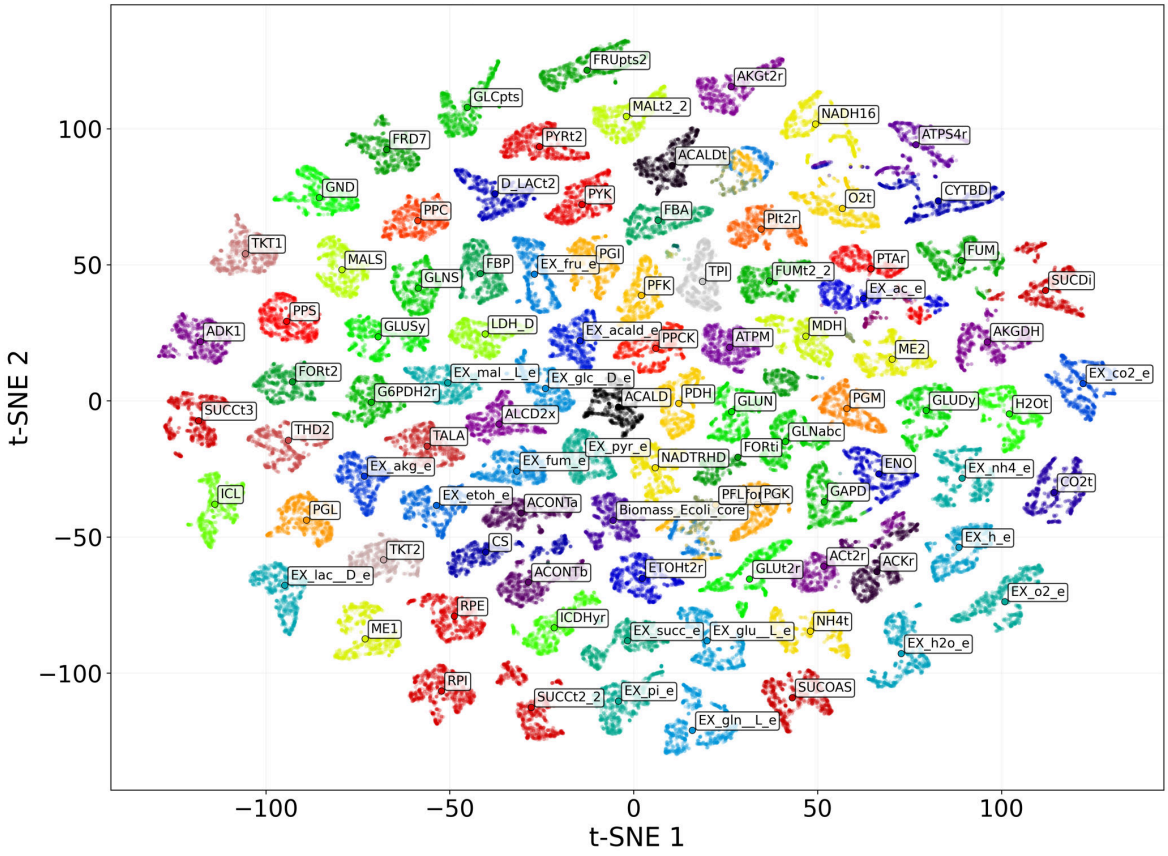


Figure 4.4: t-SNE projection of final layer FluxTransformer embeddings for the *E. coli* core model from 1000 sampled nutrient compositions. Each point represents one reaction token under one sampled condition. Colors indicate reaction identity across the 95 reactions in the core model.

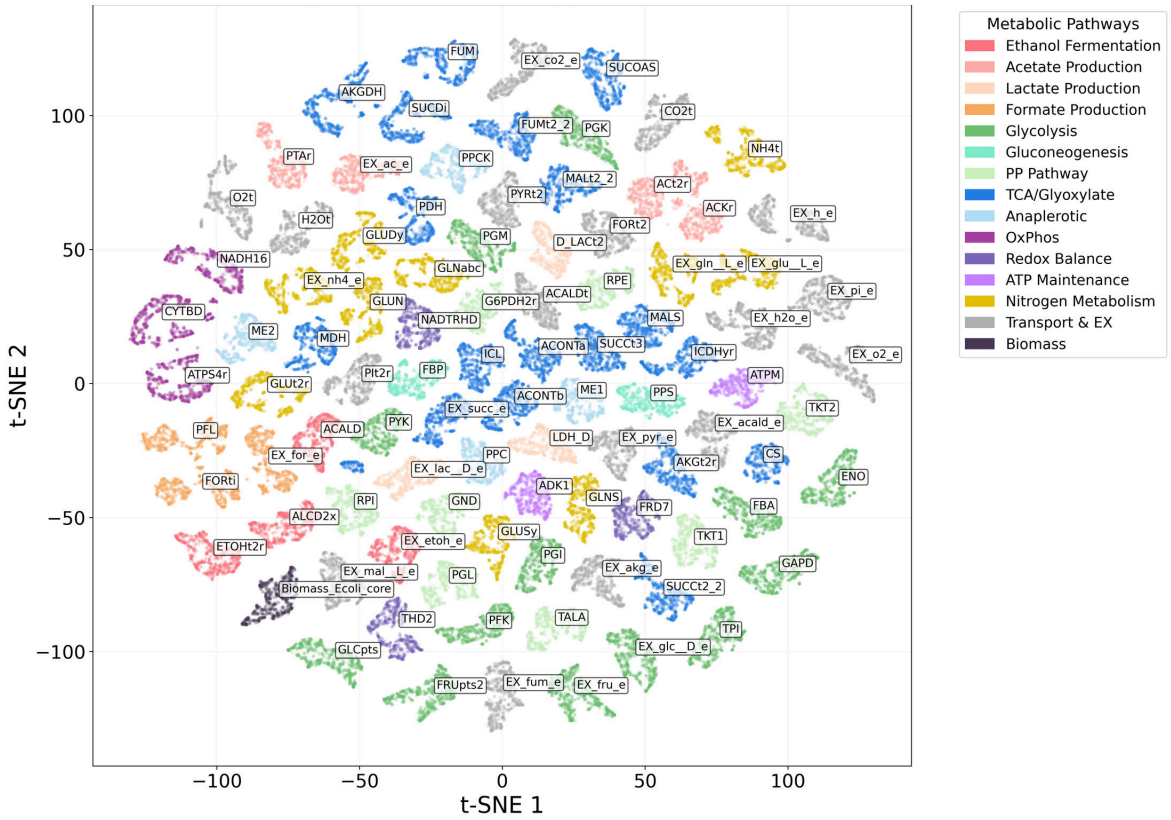


Figure 4.5: t-SNE projection of final layer FluxTransformer embeddings for the *E. coli* core model from 1000 sampled nutrient conditions. Colors indicate manually assigned pathway groups.

Figure 4.5 shows the same projection, but the colors now indicate pathway groups. Each reaction was placed into one pathway label according to its main metabolic role. This grouping is approximate because some reactions connect different pathways and could also fit under another label. Many reactions from the same pathway appear near each other, but this pattern is clearer for some pathways than for others. For example, ethanol, formate, and several glycolysis reactions appear as local groups. In contrast, TCA-cycle reactions, pentose phosphate pathway reactions, and transport and exchange reactions are more spread out. This is consistent with the connected structure of the core model, where many reactions link several parts of central metabolism.

Figure 4.6 groups the same embeddings using Spearman correlation of the FBA fluxes. Reactions receive the same color when their fluxes change in a similar way across the sampled nutrient conditions. The plot shows three large correlation groups. Increasing the number of groups beyond these three only separated individual reaction clusters. In the projection, group 0 is concentrated mainly on the right and lower regions, whereas group 2 appears mainly on the left and upper regions. Group 1 is more dispersed and overlaps with both. The projection therefore shows a partial correspondence between flux-correlation groups and embedding location.

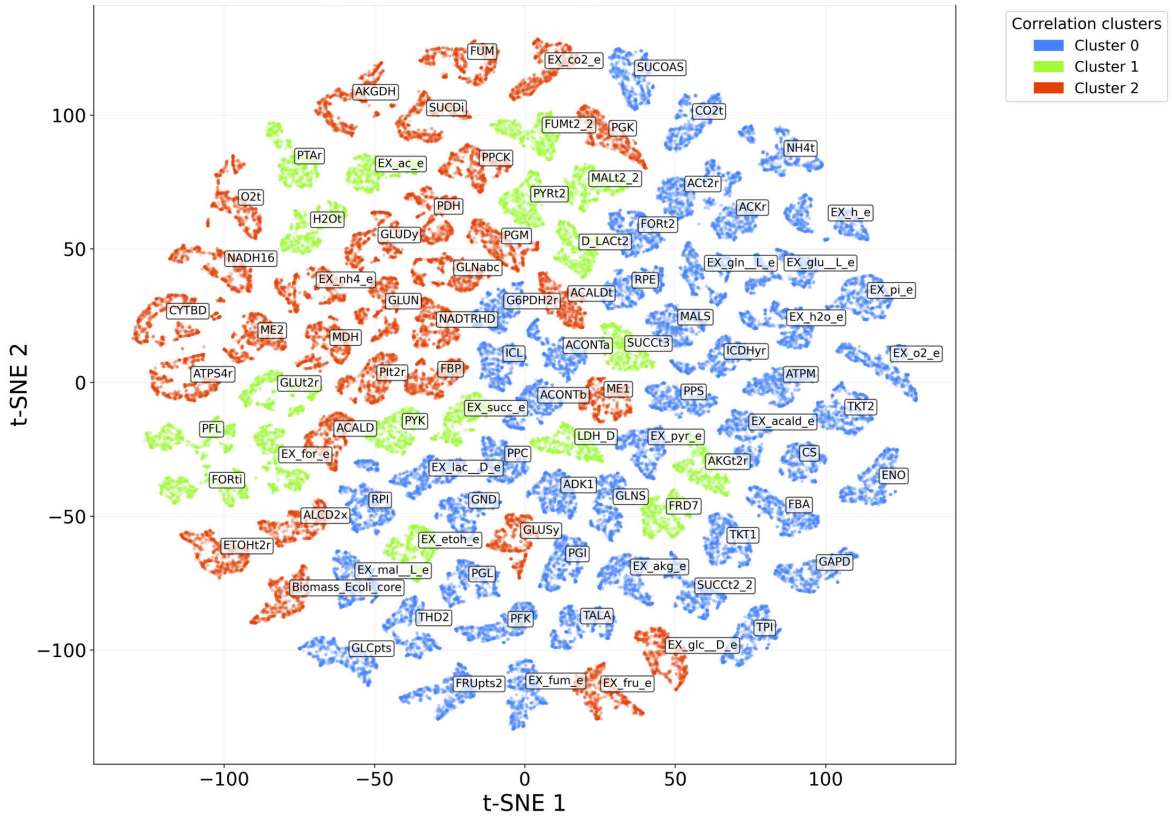


Figure 4.6: t-SNE projection of final layer FluxTransformer embeddings for the *E. coli* core model from 1000 sampled nutrient conditions. Colors indicate reaction groups based on Spearman correlation of FBA fluxes across sampled nutrient conditions.

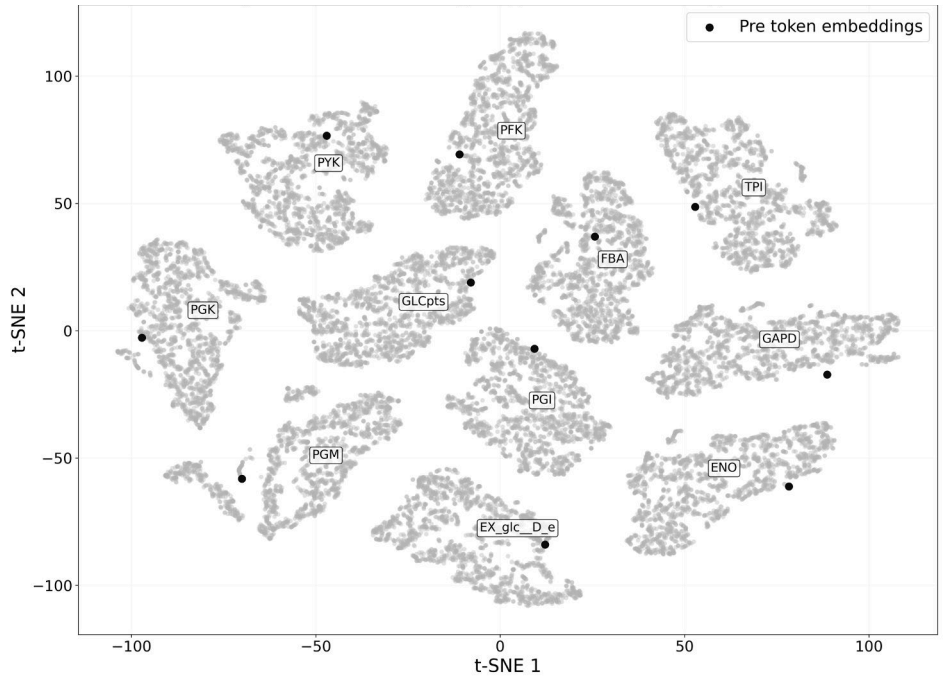


Figure 4.7: t-SNE projection of glycolysis reaction embeddings in the *E. coli* core model. Gray points show final layer embeddings across sampled nutrient compositions. Black points show the learned initial reaction embeddings before Transformer processing.

Smaller pathway-level plots were then used to examine selected parts of the embedding space. Figure 4.7 uses glycolysis as an example, because several glycolysis reactions formed a visible local group in the pathway-colored projection. The plot also compares two stages of the reaction representation. The black points represent the learned initial reaction embeddings before the nutrient condition is processed by the FluxTransformer layers. The gray regions contain the final layer embeddings across sampled nutrient compositions. Their positions relative to the black points show how the fixed initial embeddings develop into condition-specific representations as the nutrient inputs pass through the model.

Figure 4.8 uses condition-similarity coloring for two selected reaction sets. The left panel shows reactions from the pentose phosphate pathway. The right panel shows AKGDH and SUCDi, the alpha-ketoglutarate dehydrogenase and succinate dehydrogenase reactions of the TCA cycle. The rainbow scale runs from red to purple over the sampled nutrient conditions after the normalized inputs were ordered by Euclidean distance. Points with nearby colors therefore correspond to similar nutrient compositions.

In the left plot, several pentose phosphate pathway clusters show gradual changes along the color scale. This indicates that the final embeddings shift in an ordered way as the nutrient condition changes. Nearby positions within these clusters often correspond to similar nutrient conditions.

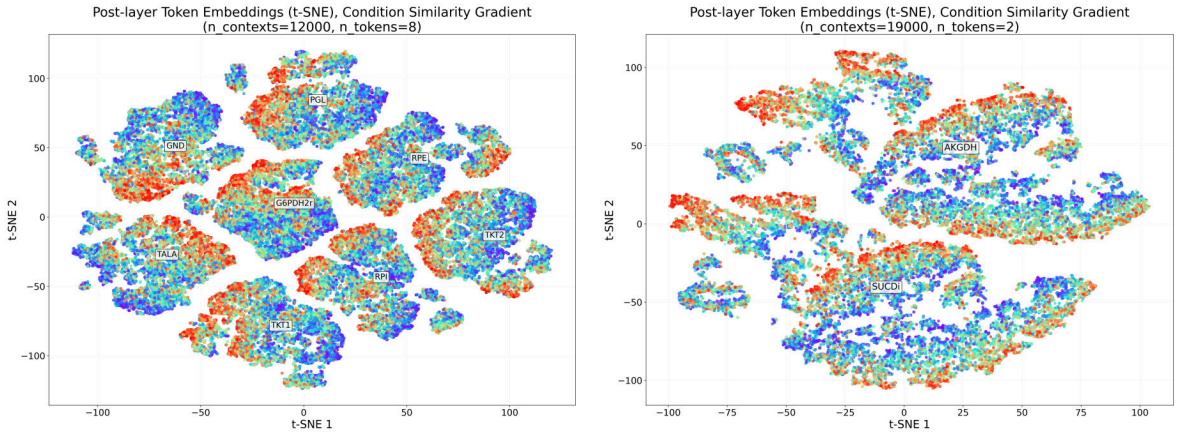


Figure 4.8: Condition-similarity coloring of selected t-SNE reaction subsets. Colors indicate sampled nutrient compositions ordered by Euclidean distance in normalized input space. **Left:** Pentose phosphate pathway reactions. **Right:** AKGDH and SUCDi reaction embeddings.

The right plot then examines this structure at the level of two individual reactions. In the full t-SNE projection in Figure 4.4, some reaction clusters appeared to have similar shapes. AKGDH and SUCDi were selected as one clear example. The two reactions remain separated, so the model still represents them as distinct reactions. Their similar shape and matching red-to-purple ordering show that the internal structure of these two clusters is also similar. This suggests that some reaction embeddings change in similar ways across nutrient conditions. This similarity is biologically plausible because AKGDH and SUCDi both operate in the TCA cycle [53].

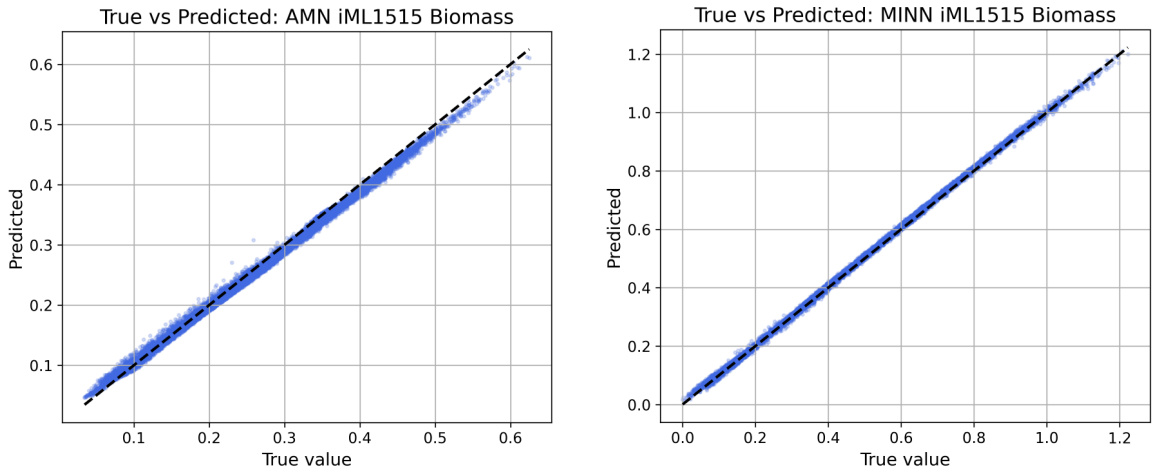
The plots show clear reaction-specific structure in the embedding space. The clearest pattern is reaction identity, as most reactions form their own clusters across nutrient conditions. Reactions from the same pathway also appear near each other in some parts of the projection. Because t-SNE is a qualitative visualization, these figures do not explain how FluxTransformer produces these embeddings or how they contribute to flux predictions.

4.2 Genome-Scale *Escherichia coli*: iML1515

4.2.1 Flux Predictions

The iML1515 experiments extend the FluxTransformer evaluation from the compact *E. coli* core model to a genome-scale reconstruction [49]. Here, the aim is to evaluate the simulated-data reservoirs used in the two experimental tasks: AMN-style growth prediction [21] and MINN-style missing-flux prediction [63].

Both iML1515 reservoirs were trained on 500,000 simulated samples, compared with 10 million samples for the *E. coli* core model, to keep genome-scale training



(a) AMN-style reservoir

(b) MINN-style reservoir

Figure 4.9: Biomass-flux predictions on simulated test data for the two iML1515 reservoirs.

computationally feasible. Figure 4.9 shows biomass-flux predictions on the simulated test data for these two models. In both panels, the predictions closely follow the identity line. The MINN-style predictions lie slightly closer to the identity line. This was expected, because the reservoir was trained on a less variable set of nutrient conditions.

Table 4.1 reports the biomass metrics and overall flux R^2 across all flux predictions in the simulated test set. Both reservoirs still predicted simulated fluxes accurately despite the genome-scale reconstruction and smaller training set.

Table 4.1: iML1515 simulated flux prediction metrics.

Reservoir	Samples	Biomass flux			Overall
		R^2	MAE	RMSE	R^2
Growth prediction	500k	0.993	0.009	0.010	0.972
Missing exchange	500k	0.999	0.008	0.009	0.999

4.2.2 Growth Prediction from Medium Composition

The frozen iML1515 FluxTransformer was next evaluated on experimental *E. coli* growth data from Faure et al. [21], using the procedure described in Section 3.5.2. The predicted biomass flux was compared with the measured mean growth rate across all 110 media. The reported results are based on out-of-fold predictions from the repeated stratified 10-fold cross-validation.

Figure 4.10 compares these results with the published AMN-QP results. The left panel reproduces Figure 3a from Faure et al. [21], and the right panel shows the out-of-fold predictions from the MLP + frozen FluxTransformer model, with one point

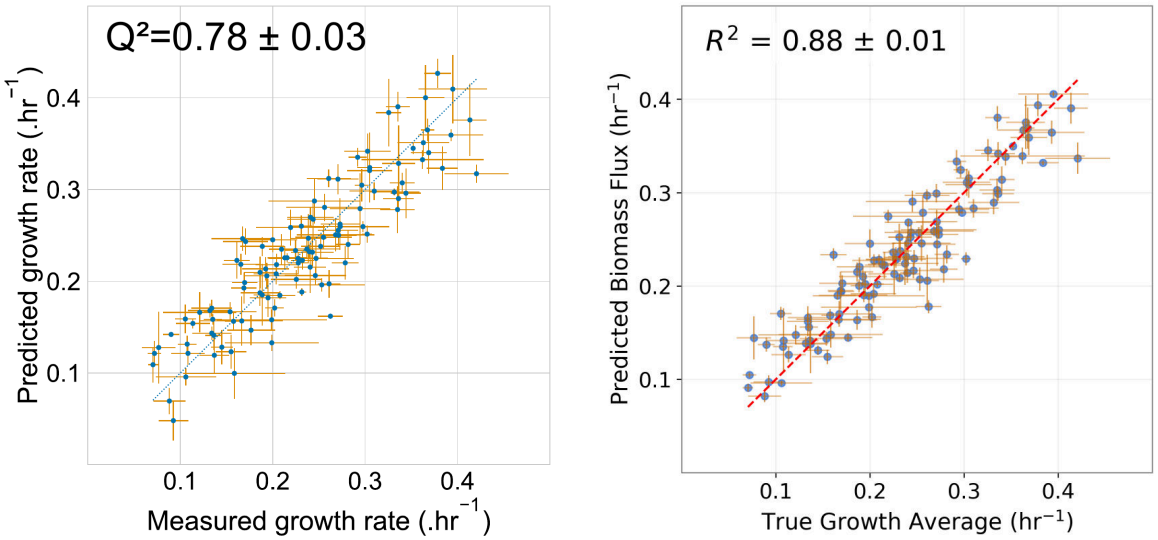


Figure 4.10: Comparison of growth prediction results on the experimental *E. coli* dataset. **Left:** AMN-QP result reproduced from Figure 3a of Faure et al. [21]. **Right:** MLP + frozen FluxTransformer model, where the MLP predicts input bounds for the frozen iML1515 FluxTransformer. Both panels show held-out predictions from three repeats of stratified 10-fold cross-validation. Horizontal error bars show measured growth-rate variability, and vertical error bars show prediction variability across repeated cross-validation runs.

per medium condition. Horizontal error bars show the standard deviation of the experimental replicates, and vertical error bars show prediction variability across repeated cross-validation runs. AMN-QP was selected as the reference because it shared the highest reported Q^2 among the AMN variants.

Table 4.2 summarizes performance across models. Faure et al. denote the held-out coefficient of determination as Q^2. The same metric is reported here as R^2, so the two are directly comparable [21]. The MLP + FluxTransformer model outperforms all three AMN variants by 0.10–0.11 and achieves the highest cross-validated score in the comparison. Its MAE is close to the average experimental growth-rate standard deviation (~ 0.02 h^{-1}). MAE and RMSE were not published for the AMN results, so they are reported only for the models evaluated in this thesis.

Faure et al. [21] compared the AMN models with a Random Forest that had been trained directly on the medium-composition features. The Random Forest achieved $Q^2 = 0.71 \pm 0.01$, a result that was lower than that of all three AMN variants. To determine whether this advantage was still present when compared to a potentially stronger black-box model, we repeated the comparison using TabPFN-3 [26], described in Section 3.6, with the same binary medium-composition features. TabPFN-3 obtained an $R^2 = 0.80 \pm 0.01$, outperforming all three AMN variants.

AMN does not turn out to be better than the more powerful TabPFN-3 baseline, which leads to the question of whether training neural proxies is actually better than

predicting directly from the medium-composition features. In contrast, the frozen FluxTransformer reservoir achieved $R^2 = 0.88 \pm 0.01$ and thus outperformed TabPFN-3. Unlike AMN, FluxTransformer therefore still retains an advantage over a stronger black-box baseline. This confirms that simulated pretraining is valuable when learning neural metabolic surrogates.

Table 4.2: Growth-prediction performance on 110 experimental *E. coli* medium conditions. The AMN results are from Faure et al. [21]; TabPFN-3 and MLP + FluxTransformer were evaluated in this thesis. MAE and RMSE are reported in h^{-1}; n.r., not reported.

Model	Q^2 / R^2	MAE	RMSE
AMN-QP	0.78 ± 0.03	n.r.	n.r.
AMN-LP	0.78 ± 0.01	n.r.	n.r.
AMN-Wt	0.77 ± 0.01	n.r.	n.r.
TabPFN-3	0.80 ± 0.01	0.03	0.04
MLP + FluxTransformer	0.88 ± 0.01	0.02	0.03

4.2.3 Prediction of Missing Exchange Fluxes

Table 4.3 reproduces the reservoir-style pFBA comparison from Tazza et al. [63] for the missing-flux experiment. The first four rows use the original MINN-fitted fluxomics file, which was prepared for the reduced iAF1260-based model setup used in the MINN-Reservoir comparison. pFBA uses glucose and oxygen uptake as input constraints, and the reservoir model predicts additional exchange constraints for CO_2, ethanol, and acetate. The first two rows report the published reference results. Adding the MINN-reservoir predictions increases R^2 from 0.892 to 0.910 and reduces MAE, RMSE, and NE [63].

The "pFBA iML1515" row shows the same pFBA baseline after replacing the reduced iAF1260-based model with iML1515. The metrics differ only marginally from the original pFBA baseline. Thus, replacing the reduced model with iML1515 had little effect on the mean metrics in this comparison. When the missing exchange constraints are predicted with the MLP + FluxTransformer reservoir and added to pFBA, R^2 increases from 0.893 to 0.908, while MAE, RMSE, and NE decrease. Its mean scores are slightly weaker than those of MINN-Reservoir + pFBA.

The final row, "FT + pFBA, iML1515 fit", uses the fluxomics file fitted separately into the iML1515 solution space. It has the highest R^2 and the lowest MAE, RMSE, and NE in the table. However, it is not directly comparable with the rows above because the target fluxomics file is different. This row reports the result when the fitted fluxomics data and the modeling pipeline use the same GEM.

Table 4.3: Performance on the missing-flux prediction task. The first four rows use the original MINN fitted fluxomics file, while the final row uses the iML1515-specific fit. Published reference results are from Tazza et al. [63]. FT denotes FluxTransformer.

Model	R^2	MAE	RMSE	NE
pFBA	0.892 ± 0.127	0.496 ± 0.353	0.836 ± 0.625	0.306 ± 0.367
MINN-Reservoir + pFBA	0.910 ± 0.091	0.445 ± 0.261	0.740 ± 0.421	0.253 ± 0.159
pFBA iML1515	0.893 ± 0.132	0.495 ± 0.366	0.832 ± 0.634	0.309 ± 0.373
FT + pFBA	0.908 ± 0.109	0.453 ± 0.316	0.779 ± 0.561	0.284 ± 0.312
FT + pFBA, iML1515 fit	0.926 ± 0.057	0.294 ± 0.227	0.630 ± 0.442	0.201 ± 0.082

4.3 Genome-Scale *Saccharomyces cerevisiae*: Yeast9

4.3.1 Flux Predictions

The Yeast9 experiment evaluated whether FluxTransformer could scale from bacterial metabolic models to Yeast9, a larger eukaryotic reconstruction [72]. FluxTransformer was trained on 250,000 simulated medium conditions to predict the full Yeast9 flux distribution.

Figure 4.11 compares predicted biomass fluxes with the corresponding FBA-simulated values in the test set. The predicted values closely followed the identity line, with $R^2 = 0.986$, MAE = 0.052, and RMSE = 0.067. The residual distribution was slightly wider than in the experiments using the smaller metabolic models.

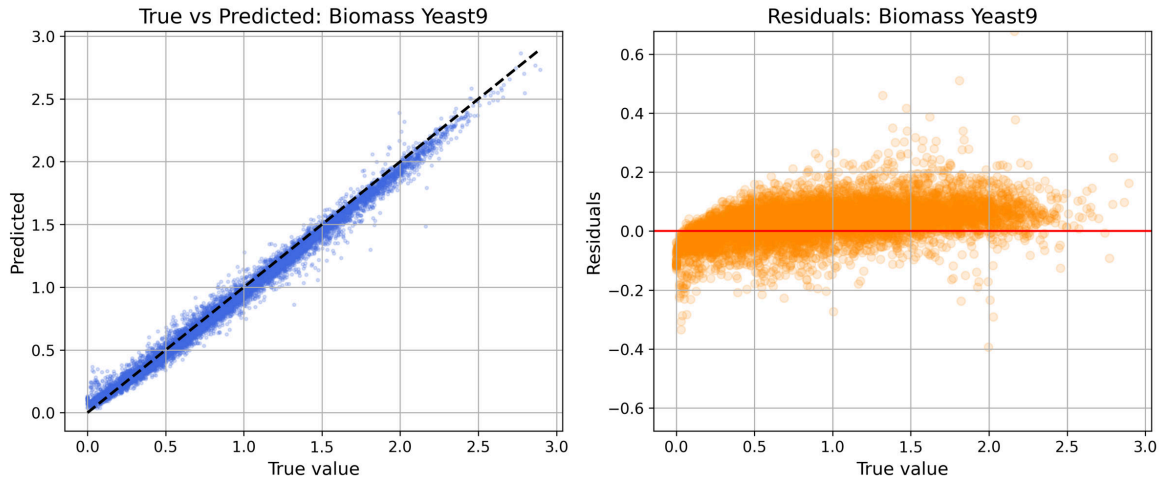


Figure 4.11: Biomass flux prediction for the *Saccharomyces cerevisiae* Yeast9 model. **Left:** predicted versus true biomass flux. **Right:** residuals (predicted minus true) plotted against the true biomass flux. The model was trained on 250 thousand simulated samples.

Pooling predictions for all 4,131 reactions resulted in $R^2 = 0.912$, an MAE of 0.065, and an RMSE of 1.002. Most of the squared error came from a small number of reactions: the five largest contributors accounted for 35.6% of the total, and the ten largest for 57.7%.

High flux did not guarantee an accurate prediction. For example, cytochrome-c oxidase and ATP synthase had R^2 values of 0.993 and 0.992, whereas phosphate transport and succinate dehydrogenase had much lower values of 0.525 and 0.431. Phosphate transport and succinate dehydrogenase were also the two largest contributors to the total squared error. The model underestimated both reactions at high flux values.

The pooled R^2 is influenced mainly by reactions with large flux variation and therefore does not describe each reaction equally. Reaction-level performance was examined by grouping reactions according to how often they were active. As shown in Table 4.4, 2,869 reactions were active in fewer than 1% of the test conditions, whereas 496 were active in more than half. Median reaction-level R^2 increased with activity frequency and was positive only for the most frequently active group.

The low-activity groups had small absolute errors because their true fluxes were usually zero or close to zero. Their R^2 values were strongly negative because the errors were measured against very little variation in the target flux. Biomass and several major fluxes were predicted well, although accuracy varied widely across the network and was poor for some high-flux and many rarely active reactions.

Table 4.4: Prediction accuracy for Yeast9 reactions with different activity frequencies in the test set. A reaction was active when $|v| > 10^{-8}$. Reactions with a true-flux standard deviation below 10^{-6} were excluded from the median R^2.

Activity frequency	Reactions	Included in median R^2	Median MAE	Median R^2
$< 1\%$	2,869 (69.5%)	1,045	0.0043	-93.865
1–10%	571 (13.8%)	562	0.0067	-1.469
10–50%	195 (4.7%)	195	0.0079	-0.331
$> 50\%$	496 (12.0%)	475	0.0181	0.540

4.3.2 Training Efficiency with Output Subsets

Training FluxTransformer on the full Yeast9 reaction set took approximately 100 hours on Puhti using 250,000 simulated samples. Figure 4.12 shows that training time increased approximately quadratically with the output subset ratio, as expected from the sequence-length dependence of self-attention.

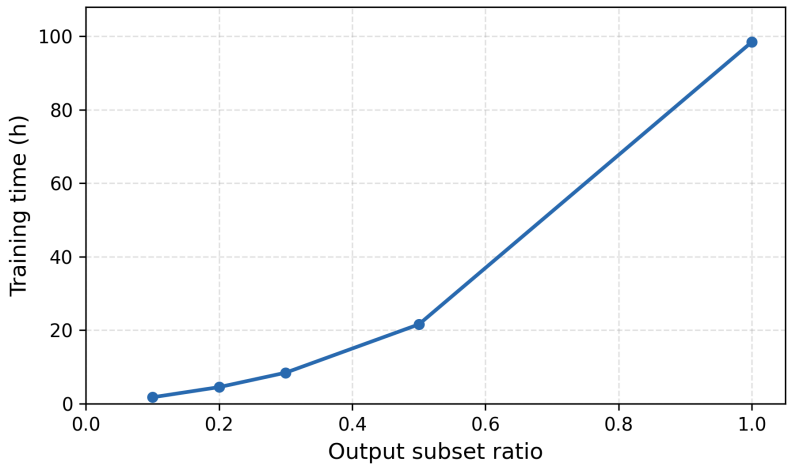


Figure 4.12: FluxTransformer training time for Yeast9 at different output subset ratios using 250,000 simulated samples.

To assess whether this reduction in training time could be achieved without a substantial loss in prediction accuracy, several preliminary output-subset strategies were tested. These experiments compared uniform, weighted, correlation-based, and sample-driven active–inactive sampling, as described in Section 3.5.4. Preliminary screening also included visual inspection of the biomass prediction plots to identify clearly unstable or poorly performing configurations. The experiments were not developed into a complete benchmark because the preliminary results showed that all tested subset strategies reduced accuracy compared with full-output training.

The full-output reference had a biomass R^2 of 0.986 and a pooled R^2 of 0.912, together with a pooled MAE of 0.065. Correlation-based sampling performed best among the subset strategies. Its strongest results used an output subset ratio of 0.5. This yielded a biomass R^2 of 0.978 and a pooled R^2 of 0.910, along with a pooled MAE of 0.075. When the subset ratio was reduced to 0.4, the biomass R^2 and pooled R^2 dropped to 0.923 and 0.905, respectively.

The other subset-selection methods were less promising in the initial experiments. Uniform sampling reduced biomass prediction accuracy, while weighting the sampling probabilities by flux magnitude improved performance over uniform sampling but remained below full-output training. Sample-driven active–inactive sampling produced low and unstable biomass prediction accuracy. Small output subset ratios of 0.1–0.3 also produced poor predictions, and extending the training time did not recover the lost accuracy.

Output-subset training reduced training time at the cost of prediction accuracy. Correlation-based sampling came closest to full-output training, especially with the subset ratio of 0.5.

5. Discussion

The chapter discusses how well FluxTransformer was able to reproduce the flux behavior generated by FBA, what insights its learned representations provided regarding the relationships between metabolic reactions, and how performance varied among the three metabolic models. It also covers transfer to experimental tasks, the limitations of the method, and the future directions for research.

5.1 Interpretation of Results

FluxTransformer was able to learn mappings from nutrient constraints to FBA- and pFBA-generated flux distributions in all studied models. It is therefore a surrogate for a defined reconstruction, objective, constraint set, and nutrient space. It does not model biological processes absent from those simulations. The findings are discussed below through the five research questions presented in Chapter 1 (Introduction).

RQ1. *Can Transformer-based neural surrogates accurately approximate FBA-based metabolic modeling?*

FluxTransformer was trained to predict the full set of reactions for each metabolic reconstruction, and no reactions were removed from the output vocabulary, which meant that the model provided a flux estimate for all reactions. With respect to the *E. coli* core model, it accurately reproduced most of the complete flux distribution, and for iML1515 and Yeast9 it was able to predict biomass as well as the main flux variations. Unlike AMN and MINN, which have explicit mechanistic layers, FluxTransformer learned these mappings using a fully neural architecture [21, 63].

Accuracy differed from reaction to reaction. Zero-inflated fluxes and reactions active in only a small fraction of the simulated conditions made rare active states difficult to learn. Reactions that reversed direction, had occasional high-magnitude values, or followed multiple metabolic regimes were also harder to predict. Their errors often appeared as false activation or compression toward common flux values. GLUN and GLUDy reactions illustrate these two failure modes. Further analysis is needed to determine whether their errors are associated with specific nutrient conditions or

metabolic regimes.

FluxTransformer also does not enforce FBA constraints during prediction. If a feasible solution is required, the predicted flux vector must be checked against the metabolic model.

RQ2. *Does the model capture biologically meaningful relationships between metabolic reactions?*

The t-SNE projection of the learned representations showed a distinct cluster for each *E. coli* core reaction. Many reactions that belonged to the same metabolic pathway appeared close to one another, and the three Spearman flux-correlation groups occupied partly different regions of the projection. In the condition-colored plots, nutrient compositions were ordered by similarity and colored from red to purple, with nearby colors representing similar conditions. The gradual color progression within clusters showed that the embeddings shifted systematically as the nutrient composition changed.

The pathway labels and the flux-correlation groups corresponded to different local structures in the projection. Because t-SNE does not preserve global distances in the two-dimensional projection [65], the plots cannot quantify these relationships in the original embedding space. They nevertheless show that the learned reaction states reflect both the reaction identity and flux co-variation.

RQ3. *How does predictive performance change across metabolic models of increasing size and complexity?*

In Yeast9, the model with the largest reaction set, reaction-level accuracy was more variable, but biomass and the dominant fluxes were still predicted accurately. A small number of reactions accounted for a large portion of the total squared error, and many reactions that were rarely active and had low variance showed a poor R^2 even though their absolute errors were small. As a result, the pooled metrics mainly reflected the dominant FBA response and concealed the weaker ones.

The longer training times for GEMs iML1515 and Yeast9 restricted the amount of architecture and hyperparameter testing that could be performed. These models also used fewer simulated conditions, while Yeast9 covered a larger nutrient space, resulting in sparser coverage of possible conditions. Increasing the number of samples would likely improve prediction performance, but would also increase training time.

RQ4. *How well does the surrogate model transfer from simulated flux data to experimental tasks?*

The most significant transfer result came from the growth-prediction experiment. The iML1515 reservoir was frozen while a small input MLP learned the mapping from experimental media to reservoir inputs. Under the same dataset and validation proto-

col, the MLP with FluxTransformer achieved higher cross-validated performance than TabPFN-3. FBA-based pretraining therefore transferred more effectively to this growth task than TabPFN-3's synthetic pretraining.

FluxTransformer also outperformed the published AMN results on the same experimental dataset under a comparable validation protocol [21]. However, this comparison is based on a single growth-prediction experiment and should not be taken as a general comparison of FluxTransformer and AMN.

The MINN-style experiment was limited to 29 conditions, which meant that the leave-one-out results were sensitive to individual samples and therefore insufficient for stable method ranking. The FluxTransformer-derived constraints still improved the adapted iML1515 pFBA baseline on the original MINN fitted targets and were close to the published MINN-reservoir result [63].

This comparison also depended on the compatibility between the fluxomics targets and the selected GEM. When the MLP + FluxTransformer + pFBA pipeline was used with the fluxomics file that had been fitted specifically for iML1515, it obtained better results than with the original MINN fitted file. This improvement mainly reflected the closer compatibility between the fitted targets and iML1515.

RQ5. *How can neural surrogate training be made more efficient for genome-scale metabolic models?*

The full-output Yeast9 training process limited the number of simulated samples and model configurations that could be evaluated. Output-subset training shortened the reaction-token sequence and decreased the computational cost. However, very small subsets omitted too much coupled reaction information. Among the tested methods, correlation-based selection gave the best trade-off between accuracy and training time. It grouped correlated reactions within the same updates and preserved more coupled flux information than the other subset rules. Although its accuracy was still below that of full-output training, it provided a compromise between speed and accuracy.

5.2 Limitations and Future Work

FluxTransformer inherits the assumptions of the FBA setup used for training and learns one selected flux distribution for each condition rather than the full set of feasible solutions. Further development is needed before it can be applied beyond the studied growth- and flux-prediction tasks. The main priorities are extending FluxTransformer to additional organisms, improving predictions for difficult reactions, and expanding the biological conditions represented in training and validation.

The simulated training and test data were drawn from the same distributions of

conditions. The reported test performance therefore does not show how well the model generalizes to constraint ranges outside the training distribution. Future evaluation should include test sets containing unseen nutrient regimes.

Future training experiments should compare activity-aware outputs, per-flux normalization, and tail-aware losses. To achieve this, targeted data generation should include situations in which difficult reactions either change their activity or attain rare high values. The methods should be evaluated across reaction groups to find out whether improving the prediction of difficult reactions leads to a decrease in accuracy for biomass or other fluxes that are well predicted. Moreover, when the predicted flux distribution must satisfy FBA constraints, constraint-aware losses or the projection of the predicted vector into the feasible solution space should also be tested.

The t-SNE analysis used for the *E. coli* core model should be extended to selected reaction groups in iML1515, since a projection of all reactions would be too dense to interpret. This would show whether the representation patterns observed in the *E. coli* core model also appear at genome scale. Embedding relationships should also be quantified in the original space and compared with pathway membership and flux correlations under controlled nutrient conditions.

Developing more general surrogates requires additional GEMs for bacteria, yeasts, and other fungi, more varied nutrient conditions, and larger simulated datasets. Automated reconstruction methods such as `CarveMe`, `gapseq`, and `CarveFungi` can be used to build organism-specific GEMs for large numbers of species [47, 73, 9]. Training surrogates for a larger number of genome-scale models is now more feasible with CSC's Roihu supercomputer, which has replaced Puhti and offers substantially more capable GPU hardware [12, 13]. Training could also include gene knockouts, reaction inhibition, and environmental factors when comparable simulated and experimental conditions are available. More efficient attention and correlation-aware output selection could further reduce the cost of training across this expanded set of conditions.

The biological targets used in this thesis also cover only part of the intended side-stream application. Biomass flux represents growth rather than microbial protein or product yield, and the FluxTransformer training conditions did not represent process factors such as pH and temperature. Therefore, accurate biomass predictions are not sufficient to show that the model can predict microbial protein production under realistic process conditions.

A general challenge in this field is the limited availability of experimental datasets that cover many nutrient conditions and include corresponding growth or flux measurements. In this thesis, the experimental evidence is limited to two small datasets of *E. coli* that were evaluated using cross-validation. To determine whether the simulation pretraining can transfer to other organisms and media, independent growth and

flux measurements are needed. Larger datasets would also allow separate test sets and more reliable performance estimates.

Future experimental datasets should cover variation in substrate composition and process conditions. They should also report growth, substrate use, major byproduct formation, and the yields of microbial protein or other valuable products. For side-stream applications, this information would allow surrogate models to be evaluated under realistic production conditions and show whether metabolic pretraining is useful beyond the current experimental settings.

A longer-term extension is a surrogate for microbial communities. Community-level metabolic models have been used to study how species compete for nutrients, exchange metabolites, and affect the relative abundance of different species within microbial communities [74, 10, 16]. These interactions matter because one species can depend on metabolites produced by another, which can affect which species are able to coexist in a community [71]. A community surrogate would need to represent organism identity, map the reactions corresponding to those organisms, and account for shared nutrient pools and cross-feeding. For side-stream applications, simulations covering many organisms could help identify suitable individual species, whereas evaluating combinations of species would require community-level simulations as well as experimental data.

6. Conclusions

This thesis developed FluxTransformer to examine whether a Transformer can learn metabolic flux predictions from FBA-generated data. The model was evaluated using the *E. coli* core model, the *E. coli* GEM iML1515, and the yeast model Yeast9. For all three models, FluxTransformer learned the relationship between nutrient conditions and the resulting metabolic state. The results show that Transformer architectures are viable neural surrogates for metabolic models, from small networks to genome-scale reconstructions.

Reaction representations remained identifiable under different nutrient conditions and reflected similarities in pathway membership and flux behavior. FluxTransformer therefore learned relationships in the metabolic network rather than predicting each reaction separately. Performance was less consistent in the larger models, but biomass and the main flux responses were predicted accurately. Full-output training gave the best results, while output-subset training reduced computational cost with some loss of accuracy.

Simulated pretraining was also useful beyond the initial FBA-prediction task. The frozen iML1515 FluxTransformer improved experimental growth prediction from medium composition compared with the reference models. It also served as a metabolic reservoir in the missing-flux experiment, although this result was more sensitive to the small dataset and the compatibility between the experimental targets and the metabolic model. These findings show that pretraining on simulated FBA data can produce reusable metabolic priors when experimental data are limited.

A natural next step is to train models for several organisms, metabolic reconstructions, nutrient environments, and genetic perturbations. Such training would test whether FluxTransformer can develop into a "universal" metabolic model that can transfer knowledge between organisms and conditions. Achieving this would require standardized reaction mappings, expanded simulated training data, and independent experimental validation. A model trained on multiple organisms could screen bacteria, yeasts, fungi, and microbial communities for circular-economy applications. It could identify promising organisms and nutrient combinations before laboratory testing and guide experiments toward efficient side-stream conversion into valuable products.

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Appendix A. FluxTransformer Architecture Experiments

Twelve FluxTransformer architectures were evaluated using the *E. coli* core model. All models were trained using 1 million FBA-simulated samples. The architectures varied in d_{model}, H, L, and d_{ff}, while dropout p_{drop} was fixed at 0.02. Table A.1 reports test-set performance over the full flux distribution, ordered by increasing overall MAE. Figure A.1 shows the corresponding biomass-flux predictions for all 12 architectures.

Table A.1: Comparison of the FluxTransformer architectures evaluated with the *E. coli* core model. Rows are ordered by increasing overall test-set MAE.

Architecture				Complexity	Validation	Test set			Training
d_{model}	H	L	d_{ff}	Params. (M)	Loss	R^2	MAE	RMSE	Time
128	8	4	1024	1.340	0.1782	0.9881	0.2467	1.2804	23:00
128	8	3	1024	1.008	0.1786	0.9880	0.2469	1.2876	27:25
256	8	3	1024	2.400	0.1787	0.9880	0.2474	1.2879	22:01
128	8	4	640	0.942	0.1784	0.9880	0.2479	1.2839	17:34
128	8	3	640	0.710	0.1788	0.9880	0.2485	1.2870	20:20
256	8	4	1024	3.192	0.1790	0.9879	0.2489	1.2888	19:28
256	8	4	640	2.401	0.1787	0.9880	0.2506	1.2836	18:10
256	8	3	640	1.806	0.1793	0.9880	0.2517	1.2870	18:22
64	8	4	640	0.409	0.1803	0.9880	0.2545	1.2836	13:21
64	8	3	640	0.309	0.1809	0.9880	0.2572	1.2838	14:03
128	8	2	640	0.477	0.1895	0.9880	0.2786	1.2881	14:51
32	4	2	640	0.098	0.2565	0.9823	0.3878	1.5599	07:48

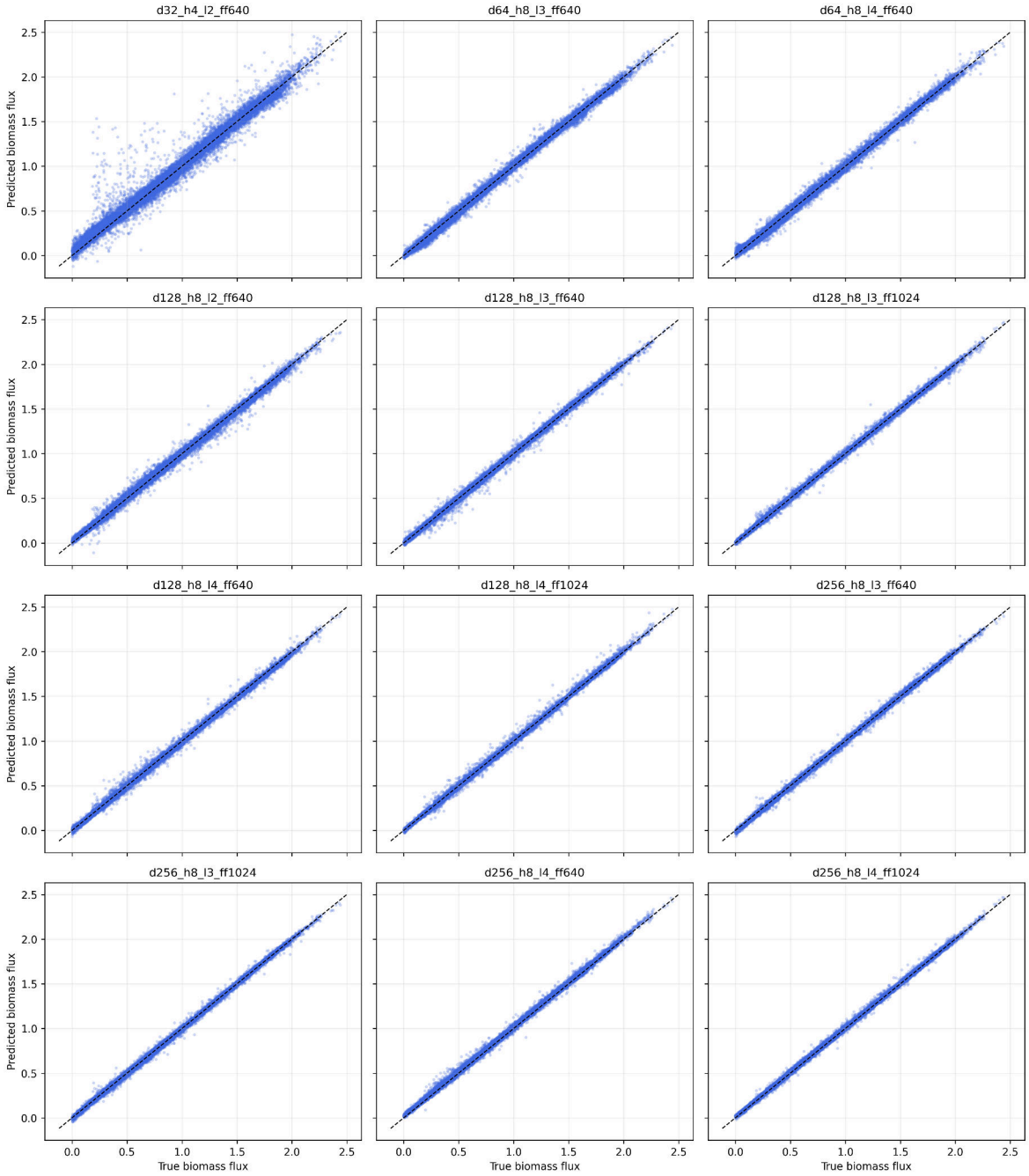


Figure A.1: Biomass-flux predictions for the 12 FluxTransformer architectures evaluated with the *E. coli* core model.

Appendix B. *E. coli* Core Model Flux Metrics

Table B.1 reports R^2, MAE, and RMSE for all 95 fluxes in the *E. coli* core model.

Table B.1: Per-flux prediction metrics for the *E. coli* core model.

Flux	R^2	MAE	RMSE	Flux	R^2	MAE	RMSE
ACALD	0.972	0.1567	0.5241	GAPD	0.990	0.1474	0.5594
ACALDt	0.988	0.0395	0.2200	GLCpts	0.992	0.0351	0.1889
ACKr	0.977	0.3418	0.9704	GLNS	0.928	0.0032	0.0357
ACONTa	0.972	0.1007	0.4176	GLNabc	0.997	0.1108	0.3438
ACONTb	0.972	0.1023	0.4177	GLUDy	0.856	1.2782	2.8951
ACt2r	0.977	0.3327	0.9695	GLUN	0.729	1.1654	2.8550
ADK1	0.812	0.0041	0.0766	GLUSy	0.119	0.9729	2.9583
AKGDH	0.995	0.2436	0.5425	GLUt2r	0.997	0.1374	0.4480
AKGt2r	0.991	0.0278	0.1877	GND	0.919	0.0403	0.2402
ALCD2x	0.964	0.1317	0.4769	H2Ot	0.974	0.5817	1.5116
ATPM	0.000	0.0056	0.0058	ICDHyr	0.970	0.1053	0.4173
ATPS4r	0.998	0.6230	1.2726	ICL	0.949	0.0034	0.0416
Biomass_Ecoli_core	0.999	0.0090	0.0120	LDH_D	0.989	0.0331	0.2177
CO2t	0.996	0.5173	1.1467	MALS	0.949	0.0033	0.0416
CS	0.971	0.1029	0.4183	MALt2_2	0.990	0.0347	0.2017
CYTBD	0.999	0.5016	1.0123	MDH	0.695	2.3887	3.4419
D_LACt2	0.989	0.0364	0.2170	ME1	-0.008	0.3362	1.5911
ENO	0.990	0.1646	0.5648	ME2	0.593	2.4100	3.4392
ETOHt2r	0.964	0.1324	0.4770	NADH16	0.998	0.4587	0.8317
EX_ac_e	0.977	0.3400	0.9667	NADTRHD	0.341	2.3442	3.4285
EX_acald_e	0.988	0.0336	0.2202	NH4t	0.994	0.2968	0.7839
EX_akg_e	0.992	0.0255	0.1869	O2t	0.999	0.2614	0.5063
EX_co2_e	0.995	0.6299	1.2153	PDH	0.966	0.3256	1.0302
EX_etoh_e	0.964	0.1289	0.4775	PFK	0.987	0.0754	0.2838
EX_for_e	0.970	0.4280	1.3417	PFL	0.970	0.4242	1.3350
EX_fru_e	0.991	0.0424	0.1991	PGI	0.979	0.0731	0.2912
EX_fum_e	0.991	0.0267	0.1921	PGK	0.990	0.1541	0.5603
EX_glc___D_e	0.992	0.0306	0.1887	PGL	0.919	0.0402	0.2393
EX_gln___L_e	0.997	0.1094	0.3431	PGM	0.990	0.1485	0.5589
EX_glu___L_e	0.997	0.1430	0.4531	PIt2r	0.999	0.0404	0.0525
EX_h_e	0.978	0.6745	1.8634	PPC	0.864	0.0031	0.0643
EX_h2o_e	0.973	0.6266	1.5338	PPCK	0.610	2.3448	3.4248
EX_lac___D_e	0.989	0.0293	0.2177	PPS	0.812	0.0039	0.0766
EX_mal___L_e	0.990	0.0313	0.2005	PTAr	0.977	0.3360	0.9674
EX_nh4_e	0.994	0.3295	0.8022	PYK	0.299	2.3428	3.4118
EX_o2_e	0.998	0.2612	0.5126	PYRt2	0.992	0.0283	0.1816
EX_pi_e	0.999	0.0439	0.0580	RPE	0.945	0.0300	0.1551
EX_pyr_e	0.992	0.0282	0.1831	RPI	0.973	0.0183	0.0782
EX_succ_e	0.927	0.0613	0.3784	SUCCt2_2	0.973	0.0255	0.1837
FBA	0.989	0.0797	0.2862	SUCCt3	0.817	0.0359	0.3417
FBP	0.998	0.0079	0.0214	SUCDi	0.994	0.2589	0.5877
FORt2	0.000	0.0006	0.0007	SUCOAS	0.995	0.2457	0.5384
FORti	0.970	0.4256	1.3364	TALA	0.929	0.0148	0.0782
FRD7	0.699	0.0199	0.2258	THD2	0.817	0.0781	0.4654
FRUpts2	0.991	0.0471	0.2010	TKT1	0.928	0.0149	0.0785
FUM	0.994	0.2884	0.6270	TKT2	0.960	0.0161	0.0773
FUMt2_2	0.991	0.0332	0.1929	TPI	0.989	0.0843	0.2898
G6PDH2r	0.920	0.0406	0.2382				